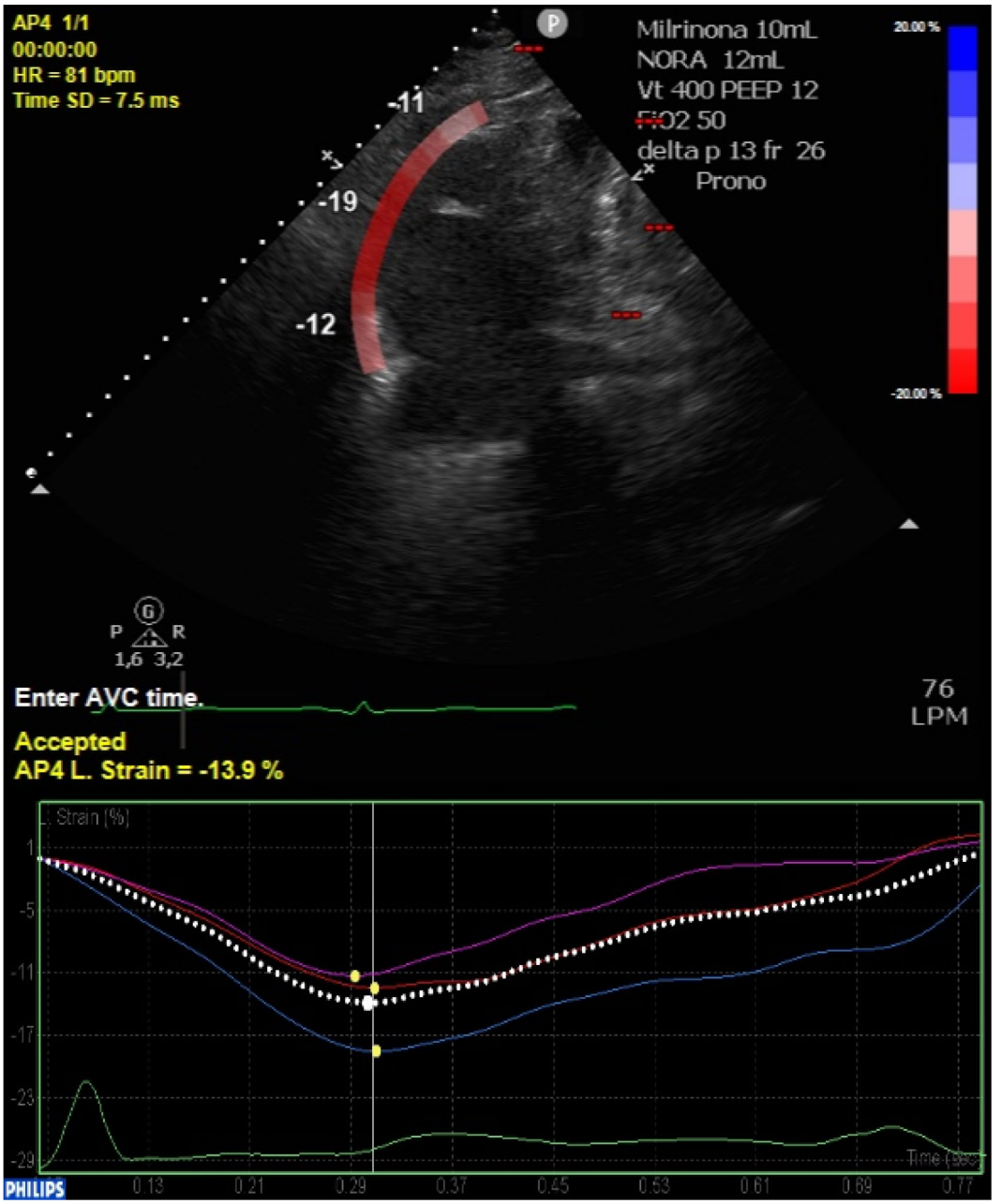


Fig. 6 Free-wall RV GLS by speckle tracking method in mechanical ventilation in prone position under vasopressor support with noradrenaline and inotropic support with milrinone showing a value of − 13.9%

deformation [10]. Because RV fibers are predominantly longitudinal, the longitudinal deformation of the free wall would adequately represent the RV systolic function [11]. It has been shown that speckle tracking of the RV free wall is a potentially better method to assess systolic function compared to traditional methods such as TAPSE, tissue S wave of the tricuspid annulus, and RV FAC [12]. A value lower than − 19% can be considered as normal [13].

On the other hand, hemodynamic evaluation of ventilated patients in prone position is of fundamental importance. For this, transesophageal echocardiography is considered as the ideal approach in critically ill patients. Nevertheless, the difficulties of this method must be taken into account, such as transesophageal probe availability, and the need for specific training. In view of this, transthoracic echocardiography has proved useful for hemodynamic assessment in

patients ventilated in prone position [4]. To this effect, the transducer is placed at the apex beat, with the mark of the probe pointed towards the left armpit, which allows a 4 and 5-chamber apical view which, in our case, allowed us to asses not only conventional parameters, but also free-wall RV GLS by speckle tracking method.

In our patient, under high-flow oxygen cannula and supine position, free-wall RV GLS value was severely compromised at − 5.6%, which slightly improved to − 8.1% with inotropic support with milrinone, not affected later by positive pressure mechanical ventilation. With the prone position maneuver, a dramatic value improvement of − 13.9% was observed, in spite of being administered the same dose of vasopressor and inotropic support.

Our case demonstrates the beneficial impact of prone position in right ventricular dysfunction not associated with ARDS, such as pulmonary thromboembolism, checked by the speckle tracking transthoracic echocardiography. The mechanisms involved could be varied. On the one hand, in prone position, heart compression force on the back regions of the lung (largely implicated in V/Q ratio) would be eliminated. This change could lower the pressure required to obtain alveolar recruitment [14]. On the other hand, by recruiting the compressed back lung, the prone position could reduce pulmonary vascular resistance and improve oxygenation, thus easing hypoxemic vasoconstriction in the lung vascular bed [6, 15].

In our case, we must not rule out the fact that consolidation of both lung bases has contributed to hypoxemia, causing some degree of pulmonary hypoxic vasoconstriction, with the resulting increased RV afterload, which got better by performing the prone position maneuver.

To our knowledge, this is the first case described of a patient diagnosed with pulmonary thromboembolism and RV dysfunction in which the prone position maneuver was carried out due to severe hypoxia, and its beneficial impact on the RV could be demonstrated using speckle tracking echocardiography.

Conclusion

Our case shows the potential beneficial effect of the prone position maneuver in severely hypoxemic patients with right ventricular dysfunction associated with pulmonary thromboembolism evidenced by speckle tracking echocardiography.

Declarations

Conflict of interest The authors have no relevant financial or non/financial interests to disclose.

Ethics statement Approval for this study was waived in accordance with the local regulations because this study is a case report of a single patient and did not include protected health information, data analysis, or testing of a hypothesis, and was de-identified.

Consent to participate Written consent was obtained from the patient before the publication of this case report.

Consent to publish The authors affirm that patient provided informed consent for publication of the images in Figs. 1, 2, 3, and 4.

Authors contributions Conceptualization: IC; methodology: IC; formal analysis and investigation: IC, LA, MVM; writing—original draft preparation: IC, LA, MVM; writing—review and editing: IC, LA, MVM, MB; supervision: GAB, MSS, RAG, PMM, FMT.

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PICTORIAL ESSAY

Tips for carotid ultrasound in the intensive care unit

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Abstract
The ultrasonography of carotid arteries plays a key role in evaluating cerebrovascular disease. There are some useful considerations to perform it correctly in the intensive care unit, such as using different kind of transducer, Doppler mode optimization, and the correct interpretation of the findings.

Keywords Point-of-care ultrasound · POCUS · Carotid ultrasonography · Carotid duplex

Introduction

Non-invasive study of carotid arteries plays a key role in evaluating cerebrovascular disease. Ultrasound is a critical tool for initial assessment by providing anatomical and hemodynamic information at the intensive care unit (ICU). Therefore, performing it correctly is of utmost importance. Some useful considerations are listed below.

Obtaining images using different transducers

Albeit high-frequency linear transducers are usually used to study carotid arteries, in certain cases like short necks, high bifurcations, and/or the presence of a central venous catheter, the use of a phased array transducer could be useful due to its reduced contact surface, as well as its capacity to reach more distal portions. In addition, transoral examination can be performed by a microconvex endocavitary transducer, which can identify the extracraneal distal portion of the internal carotid artery [1, 2] (Figs. 1, 2, 3, 4, 5, 6).

Doppler mode optimization

In color Doppler mode, in order to get a correct image using the linear transducer, the direction of the box is extremely important so that the Doppler incidence (represented by the sides of the square) has the least possible angulation in relation to the flow direction of the vessel region under study. For that, the vessel can be "tilted" by exerting greater pressure on one of the ends of the transducer and/or modifying box angulation (Figs. 7, 8).

The same principle applies to the spectral Doppler mode. It is essential to make sure the incidence between the Doppler effect (represented by a dotted line originating in the upper and middle section of the display) and the vessel direction has an angle of 60° or less. For that purpose, "the vessel can be tilted" putting pressure on one of the ends of the transducer, modifying the incidence inclination of the Doppler signal and rectifying the angle by indicating the device the actual direction of the vessel (aligning it with the dotted line with an arrow) [1, 3] (Figs. 9, 10).

Correct vessel identification

For a correct evaluation, it is essential to identify internal and external carotid arteries. In B-mode, the internal carotid artery is most of the time the most posterior one and the largest in caliber, in addition to not having visible branches at the level of the neck. Pulsed wave Doppler mode makes it possible to assess the characteristic low-resistance flow pattern of the internal carotid artery, which differs from the high-resistance pattern of the external carotid artery. This

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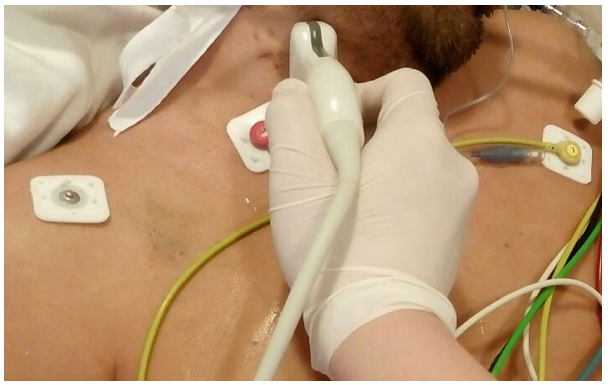


Fig. 1 Use of a linear transducer in carotid Doppler ultrasound

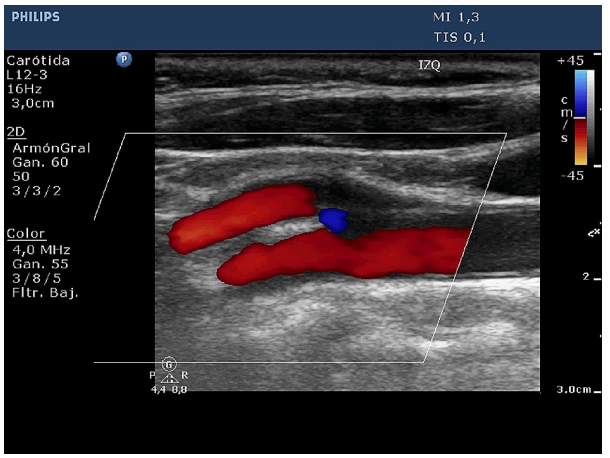


Fig. 2 Visualization of carotid bifurcation in color Doppler mode with a linear transducer



Fig. 3 Use of a phased array transducer in carotid Doppler ultrasound

can also be assessed by color Doppler, with a persistent signal even during systole and diastole in the internal carotid artery, and a more pulsatile systolic signal in the external carotid artery [1] (Figs. 11, 12, 13).

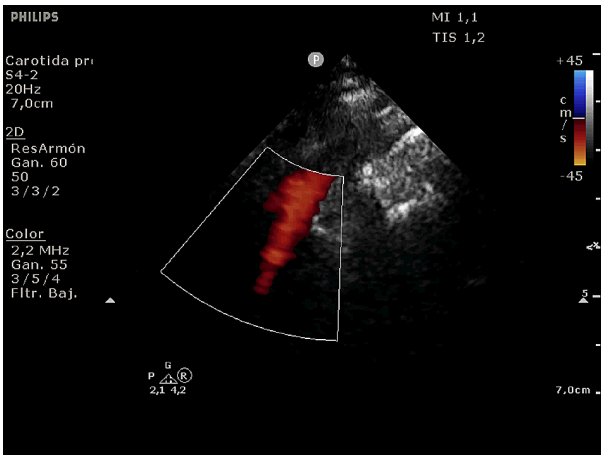


Fig. 4 Visualization of internal carotid artery in color Doppler mode with a phased array transducer



Fig. 5 Use of a transoral endocavitary transducer in carotid Doppler ultrasound

The role of Color Doppler in stenosis

Color Doppler mode enables identification of abnormal flow areas. In addition, the presence of a vessel filling defect may indicate the presence of a hypoechoic plaque not visible in B-mode. However, it is important to ensure that this is not due to an inadequate Doppler angle, or a misconfiguration of the ultrasound settings by improperly adjusting gain and/or scale. Velocity increase in stenosis usually causes a color change characterized by a mosaic of colors due to the aliasing phenomenon. Thus, imaging in color Doppler mode can help localize the stenosis area for subsequent examination with spectral Doppler [1, 4] (Figs. 14, 15).

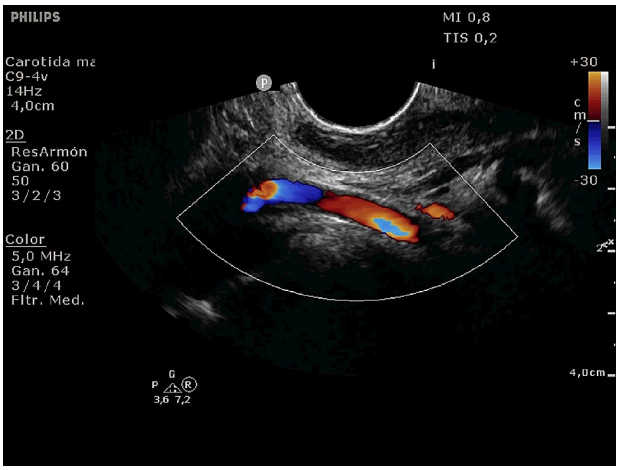


Fig. 6 Visualization of internal carotid artery in color Doppler mode with a transoral endocavitary transducer

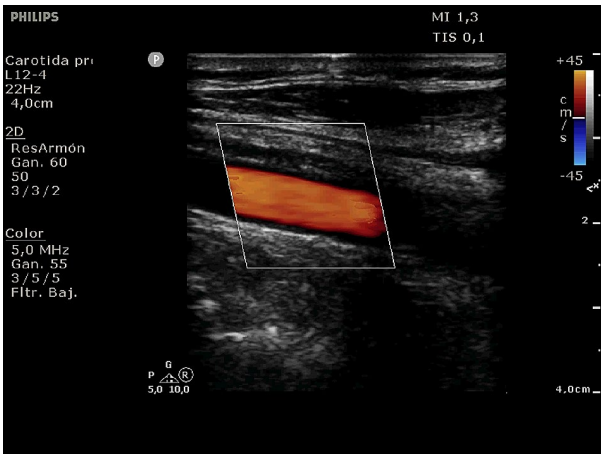


Fig. 8 Optimization of the angle among box sides and the common carotid artery "manually tilting the vessel" and modifying angulation, thus obtaining good color Doppler signal

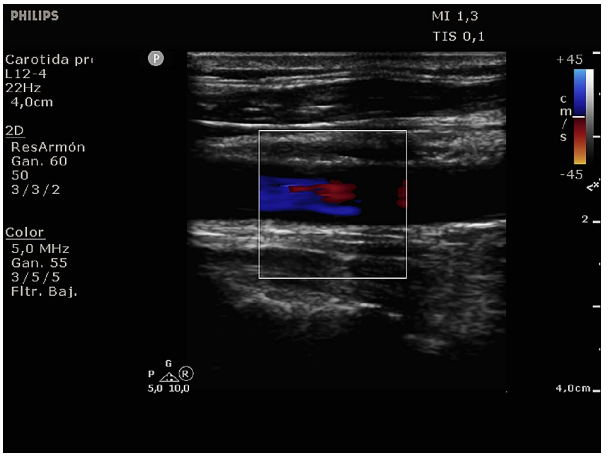


Fig. 7 Color Doppler mode at the level of the common carotid artery with box sides forming a 90° angulation in relation to the vessel with poor-signal

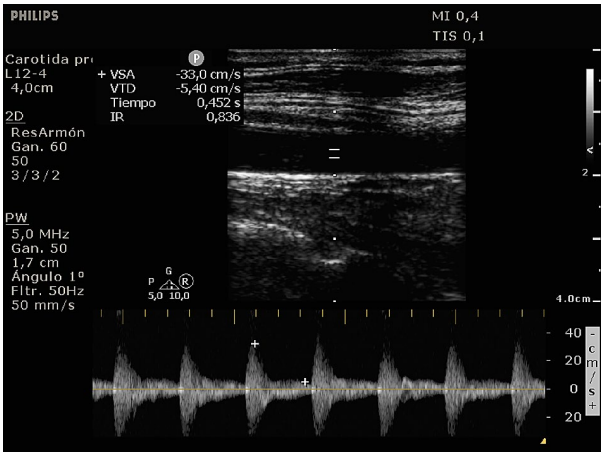


Fig. 9 Pulsed wave Doppler mode at the level of the common carotid artery with sample volume at 90° of the vessel under study, with incorrectly low peak systolic and tele-diastolic velocities

Carotid total occlusion vs. sub-occlusion

Total lack of color Doppler signal in a vessel could indicate that the vessel is occluded, but this should be confirmed by optimizing color settings to detect low-velocity flow (by decreasing scale and increasing gain) as in the case of critical stenoses. Due to its greater angle independence and greater flow sensitivity, power Doppler mode can be extremely useful in these cases, as well as in detecting complex stenoses [5] (Figs. 16, 17).

Holistic interpretation

It should be taken into account that velocity increase in a portion of the artery under study does not necessarily imply the presence of stenosis, since there are other situations in which turbulent flow can be observed with a resulting increase in velocities, as with kinking. Therefore, it is important to interpret the information obtained by spectral Doppler in the context of the information obtained with B-mode and color Doppler mode. It is also worth mentioning that velocities and flow characteristics observed in a given section of a vessel depend not only on local conditions, but also on what occurs in other areas of such vessel or in another vessel, as in the case of occlusion with collateral

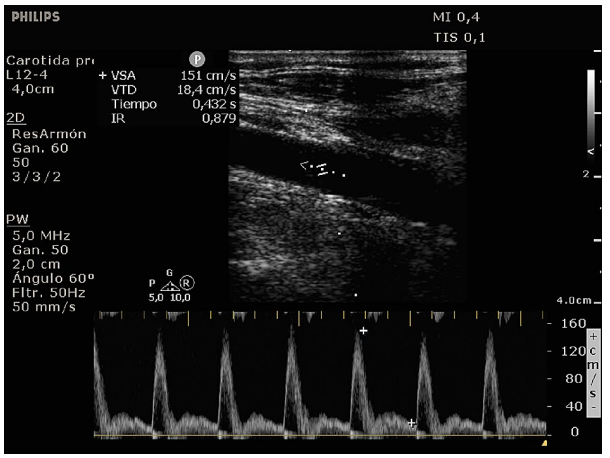


Fig. 10 Optimization of the angle between the sample volume and the direction of the common carotid artery "manually tilting the vessel", modifying signal incidence (as tilted as possible) and rectifying the angle by aligning the dotted line with an arrow with the actual direction of the artery, always ensuring a 60° angle or less, thus obtaining actual peak systolic and tele-diastolic velocity values

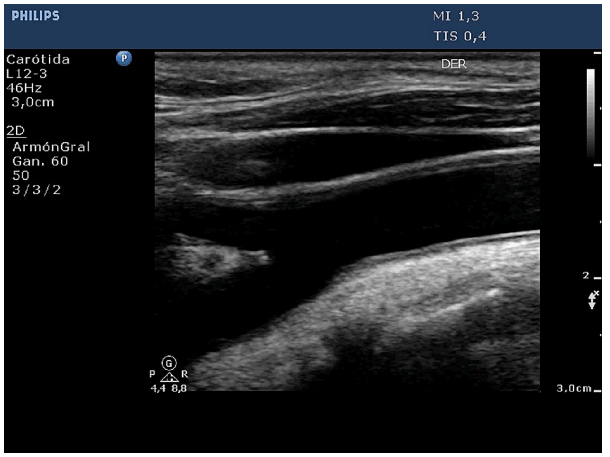


Fig. 11 B-mode showing carotid bifurcation. The external carotid artery is more anterior and smaller in caliber with respect to the ascending portion of the internal carotid artery

circulation; and on systemic conditions such as low cardiac output (where a uniform decrease in velocities in all arteries will be shown), hyperdynamics (which will manifest by a generalized increase in velocities), the presence of valve disorders as in hemodynamically significant aortic insufficiency (which will manifest through flow reversal during the entire diastole in common carotid arteries) [6–9] (Figs. 18, 19, 20, 21).

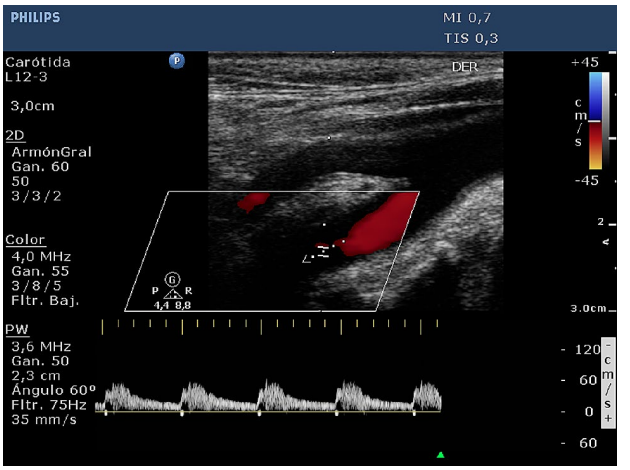


Fig. 12 Low-resistance flow pattern typical of the internal carotid artery in pulsed wave Doppler mode

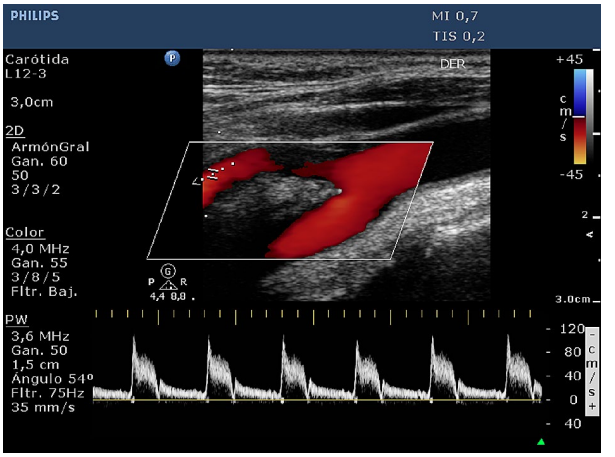


Fig. 13 High-resistance flow pattern typical of the external carotid artery in pulsed wave Doppler mode

Conclusion

Ultrasound of carotid arteries provides useful information in neurocritical care patients. For accurate diagnosis, it is important to recognize the correct technique to apply, as well as to adequately interpret the data obtained, both in B-mode and the various Doppler modes, bearing in mind the clinical context.

Authors contributions Conceptualization: IC; methodology: IC; formal analysis and investigation: IC; Writing—original draft preparation: IC; Writing—review and editing: IC, MB, GG; supervision: IC, FMT.

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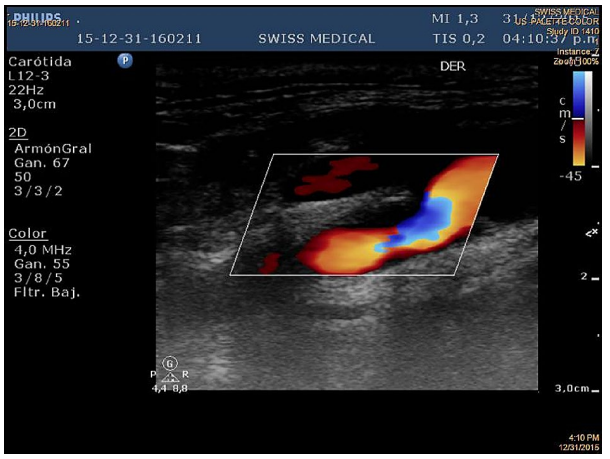


Fig. 14 Filling defect at the level of the ascending portion of the right internal carotid artery associated with the presence of sonolucent plaque, producing a mosaic color pattern (aliasing) in color Doppler mode

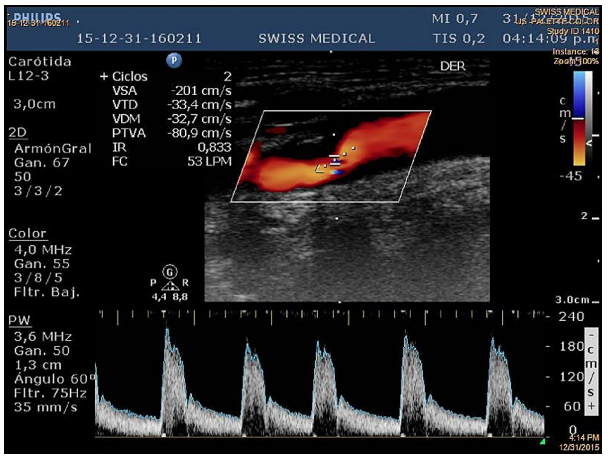


Fig. 15 Pulsed wave Doppler mode at the level of aliasing area, showing increased peak systolic velocities consistent with stenosis > 50%

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Informed consent Patients or their relatives signed informed consent regarding publishing their data and photographs.

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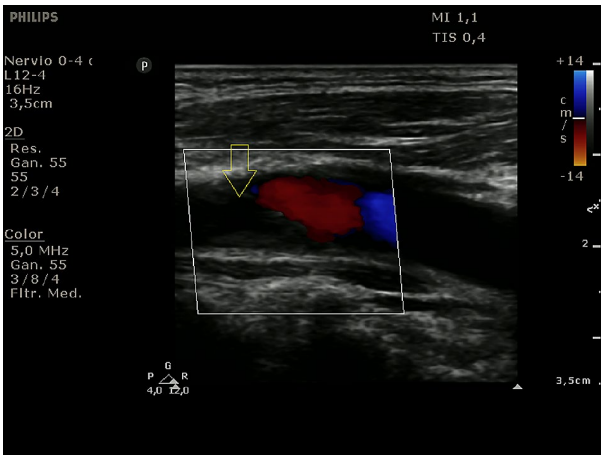


Fig. 16 Lack of color Doppler signal at the level of the ascending portion of the internal carotid artery (arrow) suggesting occlusion at that level

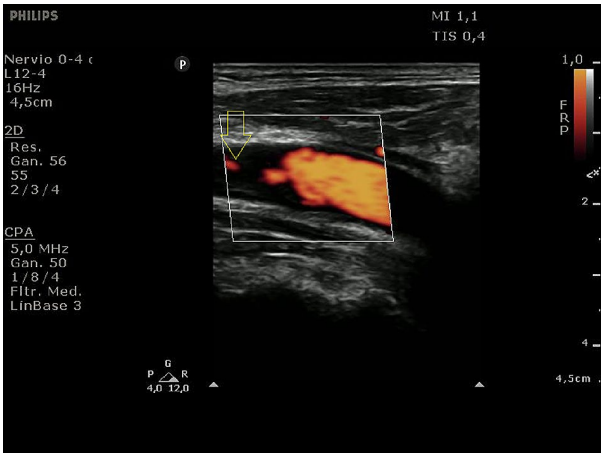


Fig. 17 Power Doppler mode shows the presence of a weak signal consistent with critical stenosis (arrow) at the level of the ascending portion of the internal carotid artery

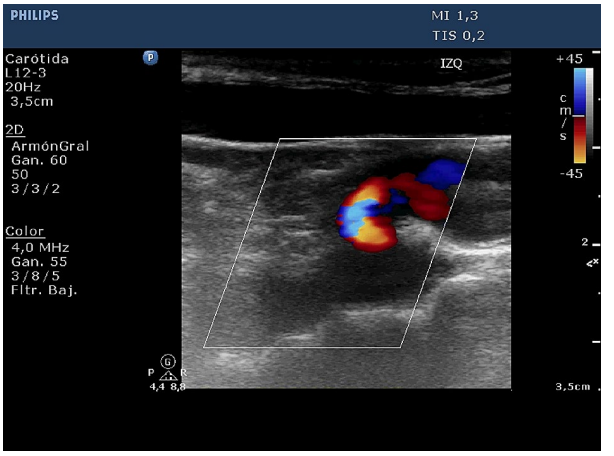


Fig. 18 Color Doppler mode showing kinking in the ascending portion of the internal carotid artery

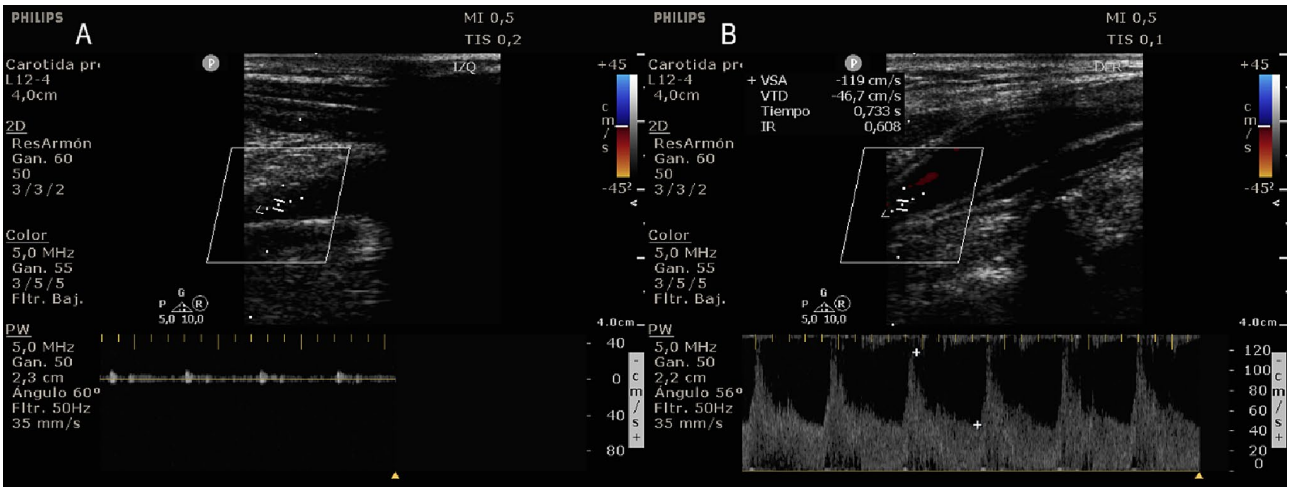


Fig. 19 Pulsed wave Doppler mode at the level of the right internal carotid artery with no flow evidence due to total occlusion, and left internal carotid artery with increased velocities due to compensatory hyperflow associated with collateral circulation at the level of the Circle of Willis

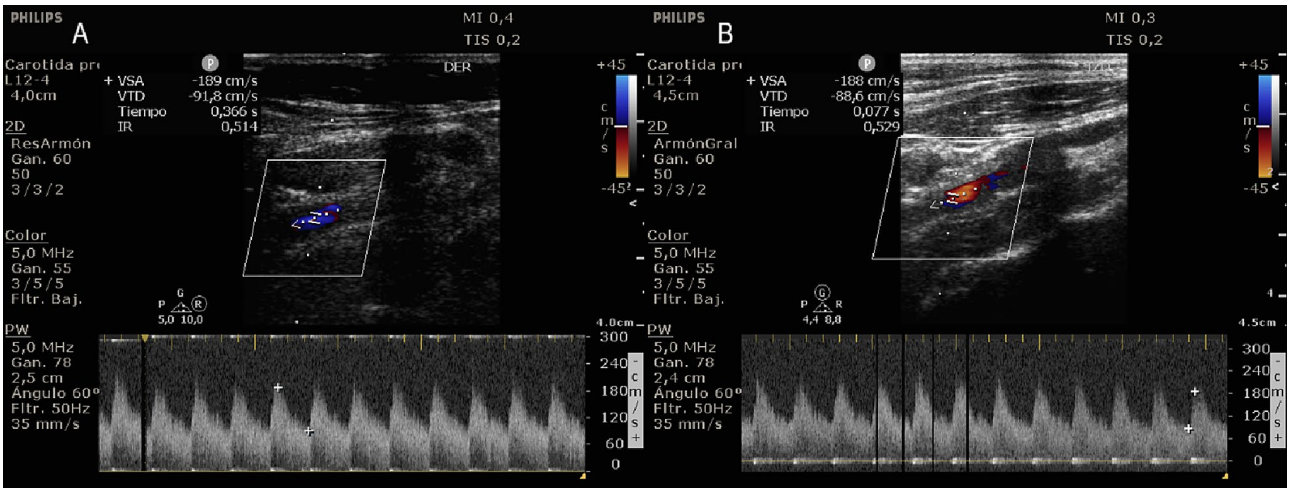


Fig. 20 Pulsed wave Doppler mode at the level of both internal carotid arteries, showing increased generalized velocities associated with hyperdynamic circulation

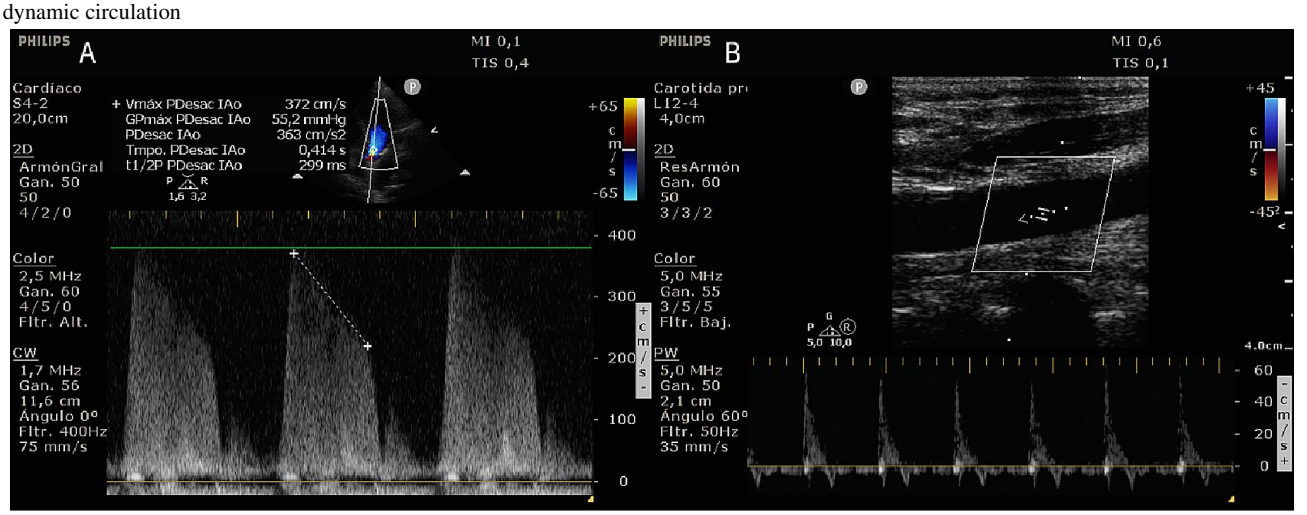


Fig. 21 **A** Continuous Doppler mode through the aortic valve in modified apical view of 5 chambers showing significant aortic regurgitation. **B** Pulsed wave Doppler mode at the level of the common carotid artery showing reversal of diastolic flow associated with aortic insufficiency

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Passive leg raising test to predict fluid responsiveness using the right ventricle outflow tract velocity–time integral through a subcostal view

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Abstract

Purpose The passive leg raising test (PLR) produces a reversible increase in venous return and, if the patient's ventricles are preload dependent, in the cardiac output. As this effect occurs in seconds, the transthoracic echocardiography is optimal for its real time assessment. The utility of the PLR for monitoring fluid responsiveness through the measurement of the left ventricle outflow tract velocity–time integral (LVOT VTI) in an apical 5-chamber view is well stablished. To achieve this view in critically ill patients is often challenging. The aim of this study is to explore the accuracy for predicting fluid responsiveness of the change in the right ventricle outflow tract velocity–time integral (RVOT VTI) from a subcostal view during a PLR.

Methods This is a diagnostic accuracy study carried out in two centers in Argentina. We included patients admitted to the intensive care unit from January 2022 to April 2022, that required fluid expansion due to signs of tissular hypoperfusion. We measured the RVOT VTI from a subcostal view in a semi-recumbent position and during the PLR, and the LVOT VTI in an apical 5-chamber view before and after a fluid bolus. If the LVOT VTI increased by 15% after the fluid bolus, the patients were considered fluid responders.

Results We included 43 patients. The area under the ROC curve for a change in the RVOT VTI during the PLR was 0.879 (95% CI 0.744–0.959). A change of 15.36% in the RVOT VTI with the PLR predicted fluid responsiveness with a sensitivity of 85.7% (95% CI 57.2%–98.2%) and specificity of 93.1% (95% CI 77.2–99.2). The positive predictive value was 85.7% (95% CI 60.8%–95.9%) and the negative predictive value was 93.1% (95% CI 78.8%–98%). The positive likelihood ratio was 12.43 and the negative predictive value was 0.15.

Conclusion The RVOT VTI change during a PLR is suitable for the prediction of fluid responsiveness in critically ill patients.

Keywords POCUS · Passive leg raising · PLR · Fluid responsiveness · Hemodynamic monitoring · Echocardiography

Introduction

The correct assessment of the volume status and the fluid responsiveness of a given patient is an everyday challenge in the intensive care unit. The latter can be evaluated by different maneuvers depending on the ventricles preload status on the Frank–Starling curve [1]. If the patient is preload dependent, the aim of a fluid bolus would be to increase the cardiac output and therefore, tissular oxygenation. Dynamic parameters are preferred for the hemodynamics assessment. The initially validated tests were on mechanically ventilated patients based on the heart–lung interactions. Among the well-known are the stroke volume (SV) variation and its surrogates, the superior vena cava collapsibility index, and the inferior vena cava distensibility index. These methods present some caveats and requirements as the presence of sinus rhythm, absence of right heart failure, tidal volume of 8 ml/kg of predicted body weight, and the absence of spontaneous breathing activity [2]. In the cases that do not fill these requirements, the passive leg raising test (PLR) can be used. This maneuver produces a reversible increase in venous return by a gravitational transfer of the blood from the lower limbs to the thorax. If the patient is preload

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dependent the augmentation in venous return will lead to a transitory increase in stroke volume. The PLR creates a reversible increase in preload that is not affected by spontaneous ventilation or arrythmias. This effect occurs in seconds thus, the stroke volume has to be assessed promptly and in real time. Hence, the transthoracic echocardiogram (TTE) is an optimal tool.

The SV can by estimated on a TTE by measuring the left ventricle outflow tract velocity–time integral (LVOT VTI) in an apical 5-chamber view [3–8]. Nevertheless, critically ill patients pose a challenge in obtaining an apical 5-chamber view due to the patient position and the mechanically ventilated air-filled lungs. The subcostal view is more feasible in these patients and allows a qualitative assessment of the heart. However, in this view the left ventricle outflow tract (LVOT) is not correctly aligned with the Doppler signal to estimate an accurate LVOT VTI [9, 10]. Instead, in the subcostal view we can visualize the right ventricle outflow tract (RVOT) perfectly aligned with the Doppler signal and therefore it is possible to obtain a right ventricle outflow tract velocity–time integral (RVOT VTI), which can be used as a surrogate of the SV as reported by Gürsel et al. [11].

The aim of this study was to assess the ability of RVOT VTI change measured by the subcostal view during the PLR to predict fluid responsiveness.

Material and methods

This is a diagnostic accuracy study, carried out in a 20 medical-surgical ICU of a tertiary-care center in Buenos Aires, Argentina (Sanatorio de Los Arcos) and an 8 medical-surgical ICU in Córdoba, Argentina (Sanatorio Privado del Interior) during a three-month period (January 2022 to April 2022). The Institutional Review Board (Comité de Ética en Investigación, Clínica y Maternidad Suizo-Argentina) approved the study procotol and all the patients gave written consent to participate.

We performed a consecutive sampling and included patients older than 18 years admitted to the intensive care unit, for whom the attending physician decided to perform fluid challenge based on signs of tissular hypoperfusion as hypotension (mean arterial pressure lower than 65 mmHg), tachycardia (more than 100 beats per minute), oliguria (urinary output lower than 20 ml/hr), lactate greater than 20 mg/dl and mottling. We excluded patients that presented inadequate quality image on the TTE or suboptimal alignment of the Doppler signal with the LVOT on an apical 5-chamber view or in the RVOT in the subcostal view, dynamic flow obstruction in the LVOT, significant aortic valve disease (stenosis and/or moderate to severe insufficiency), aortic dissection, cardiac arrythmias, abdominal hypertension, lower limbs fractures or intracranial hypertension.

Data collection

We collected the following variables from the electronic medical records: age, gender, comorbidities, diagnosis at admission, APACHE II score, mechanical ventilation and vasopressor support.

Measurements

All the patients were placed under a multiparametric monitor which recorded the heart rate, blood pressure, oxygen saturation by pulse oximetry and the electrocardiogram signal during the study. The ETT was performed by a critical echocardiography board certified operator in each center (IC, MB) with a Philips Sparq ultrasound machine (Philips Healthcare, MA, USA) with a phased array with frequency range of 2–4 MHz, and a Sonoscape SSI-6000 machine (Sonoscape medical corp., Guandong, China) with a phased array probe with frequency range of 1–6 MHz. The images were recorded and stored for offline measurements.

We performed a qualitative and quantitative assessment of the systolic function of the left ventricle, calculating the ejection fraction with the biplanar Simpson's method. We screened for valvulopathies with the color Doppler mode. The systolic function of the right ventricle was qualitatively assessed and measured with TAPSE. We measured the LVOT VTI in the apical 5- chamber view by placing the sample volume of the pulsed Doppler just above the aortic valve, while striving to maintain the angle of the Doppler signal to LVOT as close to 0°. A lineal signal corresponding to the valve closure had to be visualized right after its flow. We obtained 3 consecutively values to calculate an average. The RVOT VTI measurement was performed on a subcostal view, tilting the probe's tale down until we could visualize an image of the RVOT and the pulmonary valve. We placed the sample volume of the pulsed Doppler at the RVOT, above the pulmonary valve while achieving an angle close to 0° between de Doppler signal and the RVOT. Also, a lineal signal corresponding to the pulmonary valve closure had to be visualized following its flow. We obtained 3 consecutively values to calculate an average (Figs. 1,2).

Study protocol

The systolic arterial pressure (SAP), diastolic arterial pressure (DAP), mean arterial pressure (MAP), heart rate (HR), LVOT VTI and RVOT VTI were recorded on basal conditions with the patient's head of the bed at 45° (semi-recumbent position). Then, the PLR was performed with the patient laid flat at 0° and the lower limbs elevated at 45° with cushions. After 1 min the SAP, DAP, MAP, HR and



Fig. 1 Position of the probe for the subcostal view to visualize the RVOT

RVOT VTI were once again obtained (Fig. 3). The patient was placed back to the original position and afterwards, we infused 500 ml of Ringer's lactate solution over 15 min. A third assessment of the SAP, DAP, MAP, HR and LVOT VTI was performed. If the LVOT VTI raised more than 15% after the fluid infusion the patient was considered as fluid-responder [4]. This percentage is based on previous studies [4, 15–17]. The ventilator setting and the vasoactive drugs remained unchanged during the study protocol.

Statistical analysis

The data was analyzed with MedCalc version 20.011 (MedCalc Software Ltd, Ostend, Belgium; https://www.medcalc.org; 2021). Assuming a statistical power of 0.8 and an alpha value of 0.05 the calculated sample size was 30 patients.

The frequencies and distribution of qualitative variables were expressed in absolute quantity and percentage, while quantitative variables were expressed as means ± standard deviations or median and interquartile range, according to the observed characteristics of the distribution. The normal distribution of quantitative variables was evaluated with the Shapiro–Wilks test. The association between numeric variables was evaluated with the T student test or Wilcoxon test, depending on the distribution of the variables. The association concerning qualitative variables was studied with a Chi square test of Fisher test, based on the expected frequency of observations.

With a Hanley–McNeil test a receiver operating characteristic (ROC) curve was created to find the cut-off value for the best sensitivity, specificity, positive and negative predictive value (PPV, NPV), positive likelihood ratio (PL) and negative likelihood ratio (NL) for the percentual change of the RVOT VTI during the PLR in order to discriminate between fluid responders and non-responders [4]. For all the measurements a 95% confidence interval (95% CI) showing no overlap or a p-value lower than 0.05 was considered to be statistically significant.

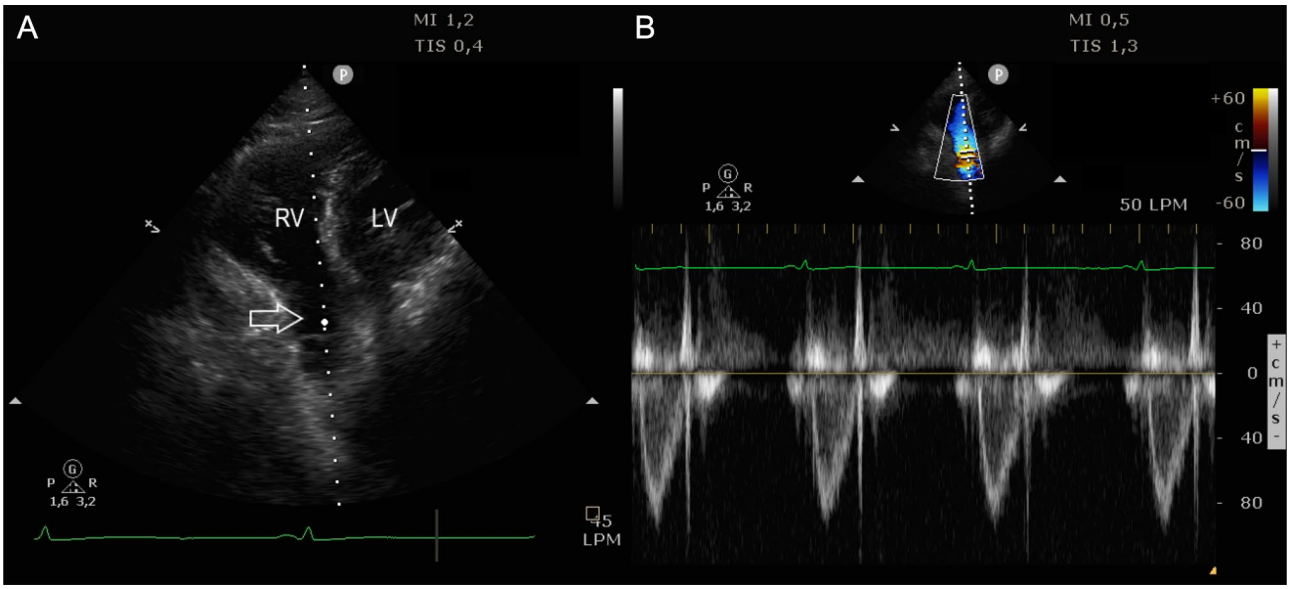


Fig. 2 **A** Subcostal view of the RVOT in B mode. Doppler signal (dotted lined) aligned with the RVOT (white arrow). **B** Pulsed wave Doppler mode at the RVOT above the pulmonary valve in the subcostal view with its flow

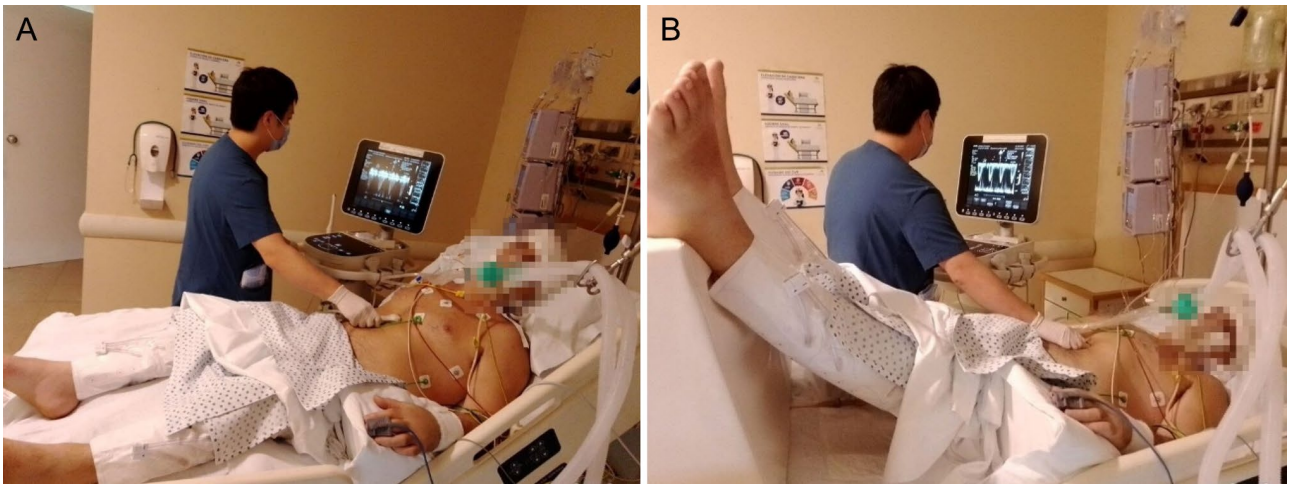


Fig. 3 RVOT VTI measurement in the subcostal view. **A** Measurement with the patient in semi-recumbent position **B** Measurement during the PLR

Results

We included a total of 43 patients between January 13, 2022 and April 25, 2022. The median age was 69 years old, with a 60.4% of men. The most frequent reason for being admitted was surgery (16.3%) and pneumonia (16.3%) followed by septic shock (11.6%). The mean of the APACHE II score was 16.9 ± 7.4. A 41.8% of the patients required mechanical ventilation and a 18.6% required vasopressors. Patients requiring mechanical ventilation were more frequently responders. Most of the patients had a preserved LV systolic function (90.4%) and RV systolic function (90.6%). Of the total of 43 patients, 14 (32.6%) were responders and 29 (67.4%) were non-responders (Table 1).

The HR, MAP and RVOT VTI showed non-statistically difference in basal conditions, and during the PLR, between responders and non-responders. Among responders, we found a significant raise in the percentual value of the RVOT VTI with the PLR ($p < 0001$) (Table 2, Fig. 4).

The area under the ROC curve for the change in the RVOT VTI induced by the PLR was 0.879 (95% CI 0.744–0.959). A change in the RVOT VTI from the subcostal view during the PLR greater than 15.36% predicted fluid responsiveness with a sensitivity of 85.7% (95% CI 57.2%–98.2%) and specificity of 93.1% (95% CI 77.2–99.2). The PPV was 85.7% (95% CI 60.8%–95.9%) and the NPV was 93.1% (95% CI 78.8%–98%). The PL was 12.43 and the NL was 0.15 (Table 3, Fig. 5).

Discussion

The change in the subcostal RVOT VTI during a PLR can predict fluid responsiveness among critically ill patients. The best cut-off value to discriminate between responders and non-responders was 15.36% with a sensitivity of 85.7% and specificity with 93.1%.

It has been established that dynamic parameters are better predictors of fluid responsiveness than static parameters. The PLR is a simple, feasible and reversible maneuver described for assessment of fluid responsiveness. The PLR creates an increase in venous return from the lower limbs to the thorax and therefore an increase in ventricular preload, simulating a reversible bolus infusion [4]. If the patient's ventricle is in a preload dependent zone of the Frank-Starling curve, the cardiac output will increase. The TTE is the optimal tool for this dynamic assessment given its cost, readiness and safety [12]. Most of the studies that assessed fluid responsiveness to PLR with TTE were based on the variation of the LVOT VTI obtained by an apical 5-chamber view. The SV was calculated multiplying the LVOT VTI value with its area, acquired from the diameter of the LVOT on a parasternal long axis view. As the area does not change in a given patient, the LVOT VTI can be used as a surrogate of the SV [10]. Multiple authors reported slightly different cut-off values for LVOT VTI variation after PLR with adequate sensitivity and specificity. Maizel et al. reported that an increase of 12% of the SV predicted fluid responsiveness with a sensitivity of 69% and a specificity of 89% in patients with suspected hypovolemia [9]. Biais et al. showed that using a cut-off value of 13% for variation of SV, the sensitivity was 100% and a specificity of 90% for predicting fluid responsiveness [4]. Préau et al. evaluated septic patients with a cut-off value of 10% with a sensitivity and specificity of 86% and 90% respectively [5]. Guinot et al. included mechanically ventilated patients under ECMO with a cut-off value for predicting fluid responsiveness of 10% with a sensitivity of 62% and a specificity of 92% [6]. Lamia et al. used the

Table 1 Demographic and clinical characteristics of the patients (n = 35)

Variable	Global $N=43$	Responders $N=14$	Non-responders $N=29$	P-value
Age (years) [1]	69 (58–75)	65 (47–74)	69 (60–76)	0.392
Male, *n* (%)	26 (60.4)	7 (50)	19 (65.5)	0.335
Diagnosis, *n* (%)				
Postoperative	7 (16.3)	1 (7.1)	6 (20.6	0.265
Pneumonia	7 (16.3)	4 (28.6)	3 (10.3)	0.132
Septic shock	5 (11.6)	2 (14.3)	3 (10.3)	0.704
Stroke	4 (9.3)	2 (14.3)	2 (6.9)	0.439
Sepsis	2 (4.7)	0	2 (6.9)	0.319
Poisoning	2 (4.7)	1 (7.1)	1 (3.5)	0.604
Subdural hematoma	2 (4.7)	0	2 (6.9)	0.319
Others	14 (32.5)	1 (7.1)	13 (44.8)	0.014
APACHE II Score[2]	16.9 (7.41)	18.1 (8.2)	16.9 (7.6)	0.637
Mechanical ventilation, *n* (%)	18 (41.86)	10 (71.4)	4 (13.8)	0.0002
Vasopressors, *n* (%)	8 (18.6)	3 (21.4)	5 (17.2)	0.742
LV systolic function, n (%)				
Normal	39 (90.6%)	13 (92.9)	26 (89.7)	0.737
Mildly impaired	2 (4.7%)	0	2 (6.9)	0.319
Severely impaired	2 (4.7%)	1 (7.1)	1 (3.5)	0.604
RV systolic function, n (%)				
Normal	39 (90.6%)	13 (92.9)	26 (89.7)	0.737
Impaired	4 (9.4%)	1 (7.1)	3 (10.3)	0.737

[1]Median (IQR)

[2] Mean (SD)

Table 2 Hemodynamic parameters at baseline and after PLR in responders and non-responders

	Responders	Non-responders	*p*-value
HR (beats/min)[1]			
Baseline	85 (25)	82 (18)	0.738
During PLR	82 (24)	82 (18)	0.955
SAP (mmHg)[1]			
Baseline	119 (19)	122 (38)	0.647
During PLR	130 (21)	127 (24)	0.742
DAP (mmHg)[1]			
Baseline	62 (15)	67 (17)	0.357
During PLR	70 (15)	70 (14)	0.885
MAP (mmHg)[1]			
Baseline	78 (17)	81 (20)	0.579
During PLR	87 (15)	84 (16)	0.652
RVOT VTI (cm)[1]			
Baseline	14.7 (3)	16.2 (4.2)	0.231
During PLR	18.2 (3.7)	17.6 (4.2)	0.643
Delta RVOT VTI with PLR (%)	22.2 (8,6)	8.6 (6.4)	< 0.0001

[1]Mean (SD). *HR* heart rate, *SAP* systolic arterial pressure, *DAP* diastolic arterial pressure, *MAP* mean arterial pressure, *RVOT VTI* right ventricular outflow tract velocity–time integral, *PLR* passive leg raising test

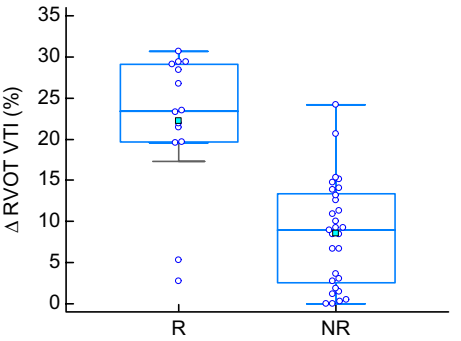


Fig. 4 Boxplot and individual values of the percentual change of the RVOT VTI during the PLR among responders (R) and non-responders (NR)

LVOT VTI value without the SV calculation obtaining a sensitivity of 77% and a specificity of 100% for a cut-off value of 12.5%. Brun et al. obtained similar results [7, 8].

As previously mentioned, the main limitation in obtaining a LVOT VTI value in critically ill patients is achieving an adequate apical 5-chamber view [12]. The subxiphoid view is more accessible in these circumstances [11]. Nevertheless, the anatomical disposition of the cardiac structures does not allow to align the Doppler signal with the LVOT but it does perfectly with the one of the RVOT [9, 11].

Table 3 Accuracy of RVOT VTI for the PLR to predict fluid responsiveness

Statistics	Value
Cutoff, %	15.36
Area under the ROC curve	0.879 (95% CI 0.744–0.959)
Sensitivity, %	85.7 (95% CI 57.2–98.2)
Specificity, %	93.1 (95% CI 77.2–99.2)
Positive predictive value	85.7% (95% CI 60.8%–95.9%)
Negative predictive value	93.1% (95% CI 78.8%–98%)
Positive likelihood ratio	12.43
Negative likelihood ratio	0.15

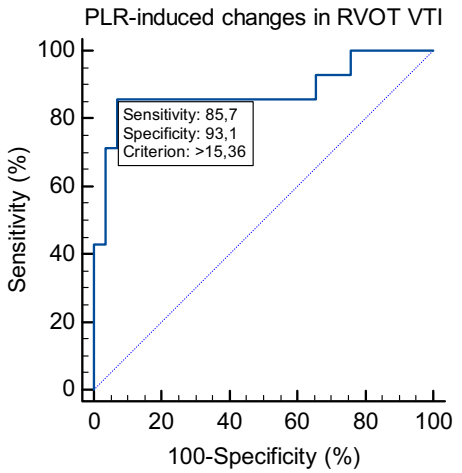


Fig. 5 ROC curve for the change in RVOT VTI during the PLR

The reliability of the PLR depends on the accuracy of the device used to measure the cardiac output; thus, the fluid responsiveness cut-off value must take in consideration this variability and has to be higher than the least significant change (LSC) of the cardiac output measurement method [13]. For echocardiography, the LSC corresponding to the VTI measurement was 5% according to the literature [13, 14]. Our proposed cut off value is substantially higher (15.4%).

This study presents several limitations. The TTE studies were performed by a single operator in each center that was not blind for the findings. The inter and intraobserver variability was not assessed. We used the cut-off value of 15% for variation in LVOT VTI after the infusion of a fluid bolus to define fluid responder, which was chosen by reference to previous studies and not assessed in this present study [4, 15–17].

To our knowledge, this is the first study that assess the PLR accuracy for predicting fluid responsiveness with the RVOT VTI obtained in a subcostal view.

Conclusion

The RVOT VTI variation through a subcostal view during the PLR is an accurate predictor of fluid responsiveness in critically ill patients.

Author contributions conceptualization: IC; methodology: IC; formal analysis and investigation: IC, VOC, MB, MFF; writing—original draft preparation: IC; writing—review and editing: IC, VOC; supervision: IC, PMM, FMT.

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Declarations

Conflict of interest The authors have no relevant financial or non/financial interests to disclose.

Ethical approval The study was conducted following the principles of the Helsinki Declaration and within the precautions established by ethical and legal standards. The study was approved by the Institutional Review Board and Ethic Committee of Clínica y Maternidad Suizo-Argentina (approval number 6517).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish The authors affirm that human research participants provided informed consent for publication of the images in Figs. 1, 2, and 3.

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CASE IMAGE IN CARDIOVASCULAR ULTRASOUND

Lung consolidation as an acoustic window to superior vena cava by transthoracic ultrasonography in the intensive care unit

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The superior vena cava (SVC), due to its anatomical location, is challenging to evaluate by transthoracic ultrasonography. There are descriptions in the literature of its evaluation through a right supraclavicular and a modified left parasternal approach, but in both cases its visualization is partial [1, 2]. We describe an unusual case in which it was possible to visualize the entire length of the SVC due to the presence of right lung consolidation.

A 56-year-old male with a medical history of arterial hypertension and obesity was admitted to the intensive care unit (ICU) due to severe Covid-19 pneumonia, requiring orotracheal intubation and mechanical ventilation support in the prone position. After several days, he presented improvement in oxygenation, for which he began the weaning process. The patient evolves with an episode of desaturation with a higher oxygen requirement. As ultrasonography has emerged as an important diagnostic tool in the Covid-19 pandemic, we performed a lung evaluation with point-of-care ultrasound (POCUS), using the phase-array transducer at 2–4 MHz, with its mark pointing towards the head, resting on the second right intercostal space at the level of the midclavicular line, with the ultrasound machine with the cardiac preset. It showed in B mode with the tissue harmonic imaging function activated, the presence of lung consolidation defined as a tissue-like pattern visible at the chest wall, arising from the pleural line [3], which served as an acoustic window that allowed visualization of the superior vena cava in its long axis flowing into the right atrium. In addition, by rotating the transducer 90º with its mark pointing to the left of the patient, an axial view of the SVC could be obtained (Fig. 1). Due to the sudden event of hypoxemia associated with the absence of fever and respiratory secretions, the condition was interpreted as lung atelectasis at the right middle lobe. A chest X-ray was requested, which confirmed the finding. Chest physiotherapy was indicated with improvement of his clinical condition. A new ultrasonographic evaluation showed the disappearance of the consolidation and no visualization of the SVC. He was later extubated and discharged from the ICU.

Under normal conditions, the presence of air in the lung does not allow visualization of the structures below the pleura. However, lung consolidation can act as an acoustic window by losing normal aeration, allowing the identification of mediastinal structures [4].

In our patient, the lung consolidation at the right middle lobe with tissue-like pattern functioned as acoustic windows that allowed clear visualization of the SVC in its entirety. This case shows that the presence of a consolidated right middle lobe may allow the evaluation of the SVC by transthoracic ultrasonography.

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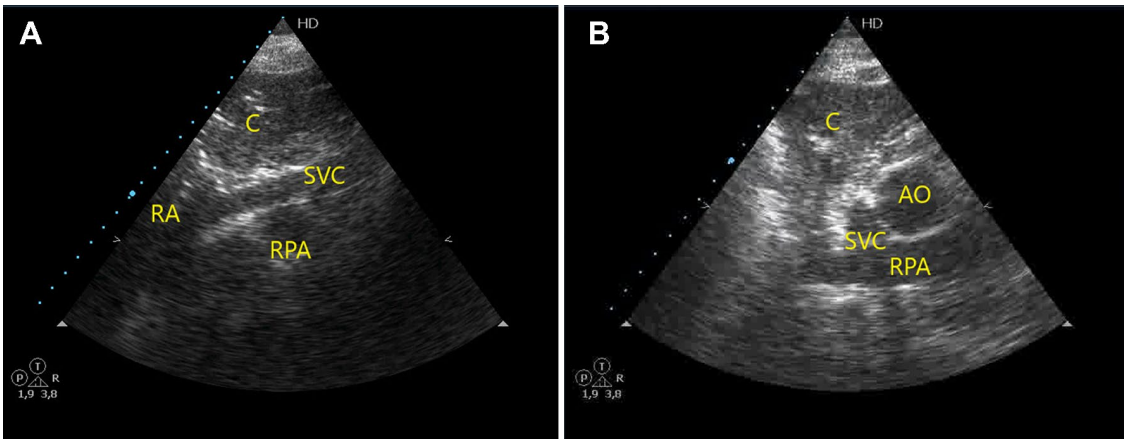


Fig. 1 Lung ultrasound at the right middle lobe. (**A**) A long axis view showing consolidation (C), superior vena cava in its long axis (SVC), right pulmonary artery (RPA), and right atrium (RA). (**B**) A Short axis view showing consolidation (C), superior vena cava in its short axis, right pulmonary artery (RPA), and aorta (AO)

Authors contributions Conceptualization: IC; Methodology: IC; Formal analysis and investigation: IC; Writing—original draft preparation: IC; Writing—review and editing: IC, CB; Supervision: FMT, PMM.

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Data availability The data that support the findings of this study are available from the corresponding author, IC, upon reasonable request.

Declarations

Conflicts of interest The author declares that he has no conflict of interest.

Written consent Written consent was obtained from the patient before the publication of this case.

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ORIGINAL PAPER

Transthoracic echocardiography of patients in prone position ventilation during the COVID-19 pandemic: an observational and retrospective study

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Abstract
Mechanical ventilation in prone position is a strategy that increases oxygenation and reduces mortality in severe ARDS. The hemodynamic and cardiovascular assessment of these patients is essential. Transthoracic echocardiography (TTE) is a widely used tool to assess hemodynamics in critical care, but the prone position is thought to limit adequate TTE views and goal-oriented measurements. The aim of this study is to show the feasibility of the hemodynamic assessment by transthoracic echocardiography during prone position ventilation (PPV). This is a retrospective, observational study, carried out in the intensive care unit (ICU) of a tertiary-care center in Buenos Aires, Argentina. We included all the adult patients admitted to the ICU between March 2020 and August 2021 who had a TTE examination in PPV due to ARDS. During the study period, we evaluated by TTE a total of 35 patients requiring PPV. The vast majority of the patients had COVID-19 pneumonia (91.4%). In 33 out of 35 (94.3%) cases, it was able to achieve an adequate apical four chamber view. We assessed qualitatively the systolic function of left ventricle (LV) and right ventricle (RV) in all of the successfully evaluated patients. We measured the RV basal diameter (94.3%), RV/LV ratio (77.1%), tricuspid annular plane systolic excursion (TAPSE) (91.4%), and septal mitral annular plane systolic excursion (MAPSE) (88.5%) in most of them. Also, we quantified the left ventricle outflow tract velocity time integral (LVOT VTI) in a large part (68.5%) of the examinations. Transthoracic echocardiography is a useful tool for the hemodynamic assessment of patients in prone position under mechanical ventilation.

Keywords Point-of-care ultrasound · POCUS · Echocardiography · Prone position · COVID

Introduction

Prone position ventilation (PPV) improves oxygenation and reduces mortality in severe or moderate-to-severe acute respiratory distress syndrome (ARDS) [1]. The survival benefit is likely related to a more homogeneous lung ventilation and reduced ventilator-induced lung injury (VILI). Current guidelines recommend prone positioning for a prolonged period of time in adult patients with ARDS and deep hypoxemia [2, 3]. Despite the evidence, the adoption of this maneuver in daily practice used to be low [4, 5], but in recent years it has been widely adopted due to the COVID-19 related ARDS (C-ARDS) [6–8].

The hemodynamic assessment of prone patients is crucial for everyday practice but may be challenging. It can be performed with right heart catheterization, transpulmonary thermodilution measurements, or echocardiography. Critical care echocardiography (CCE) is a well-established tool for cardiac function and volume status assessment [9]. Transthoracic echocardiography (TTE) is the most common approach in the critical care setting but, transesophageal echocardiography (TEE) has been advocated during prone position in ARDS patients in terms of image quality and safety [10]. Several limitations arise with the transesophageal route, including the availability of the TEE probe, costs, and specific training [9]. Moreover, early in the COVID-19 pandemic, there

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were concerns regarding the echocardiographer's safety during the TEE examination [11].

In the clinical context of C-ARDS, with a large population of mechanically ventilated patients in the prone position, the feasibility of performing TTE as a routine assessment has a crescent interest [12–14].

In this study, we reported our experience with TTE as a pragmatic, goal-oriented approach in PPV during the COVID-19 pandemic.

Materials and methods

This is a retrospective, observational study, carried out in the 20 medical-surgical bed ICU of a tertiary-care center in Buenos Aires, Argentina (Sanatorio de los Arcos), between March 2020 and August 2021. The Institutional Review Board (Comité de Ética en Investigación, Clínica y Maternidad Suizo-Argentina) approved the study protocol (approval number 6518) and waived the requirement for informed consent.

Study population

We included all the adult (≥ 18 years old) patients admitted to the ICU between March 2020 and August 2021 who had a TTE examination in PPV due to ARDS.

ARDS was diagnosed according to the Berlin definition [15]. Prone ventilation was strongly recommended in patients who met the PROSEVA criteria [1].

Intervention

As this is a pragmatic study carried out in the COVID-19 pandemic with work overload and staff shortening, the decision to perform a TTE examination in prone position was at physician discretion. We also attempted to reduce the exposition time inside the patient's room; hence the main objective was to perform a goal-oriented TTE examination.

We used a Philips Envisor HD or Sparq ultrasound machine (Philips Healthcare, Bothell, WA, USA) with a phased array S2-4 probe. All the studies were performed by a board-certified intensivist in critical care ultrasonography and echocardiography (IC).

All the patients were in the "swimmer" position. We did not make any specific adjustments to the patient bed or position. The ultrasound machine was located to the left of the prone patient. The probe was placed at the 5th–6th intercostal space, at the apex beat, between the left middle clavicular line and left anterior axillary line. The mark of the probe pointed towards cephalic and dorsal until it was possible to visualize in B mode an apical four chamber view. The five apical chambers view was obtained by slightly tilting the transducer's tail towards the dorsal region (Figs. 1, 2).

The primary objective was to perform a qualitative assessment of the LV and RV function. In all cases, an attempt was made to measure the LV and RV basal diameters and its ratio, the mitral annular plane systolic excursion (MAPSE), and the tricuspid annular plane systolic excursion (TAPSE). According to individual situations, we additionally measured transmitral E and A wave, lateral and septal mitral annulus e′ wave, E/e′ ratio, and the S′ wave of the lateral tricuspid annulus. On a case-by-case basis, if a tricuspid regurgitation flow could be identified by color Doppler, we

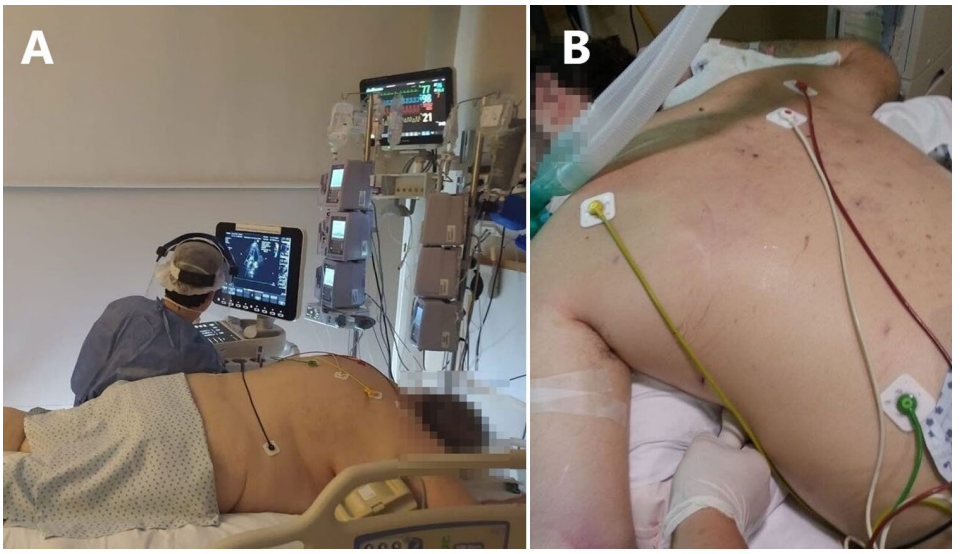


Fig. 1 **A** Transthoracic echocardiography in a patient under mechanical ventilation in prone position (patient 1). **B** Position of the transducer at the point of apex beat (between the middle clavicular line and the left anterior axillary line) (patient 2)

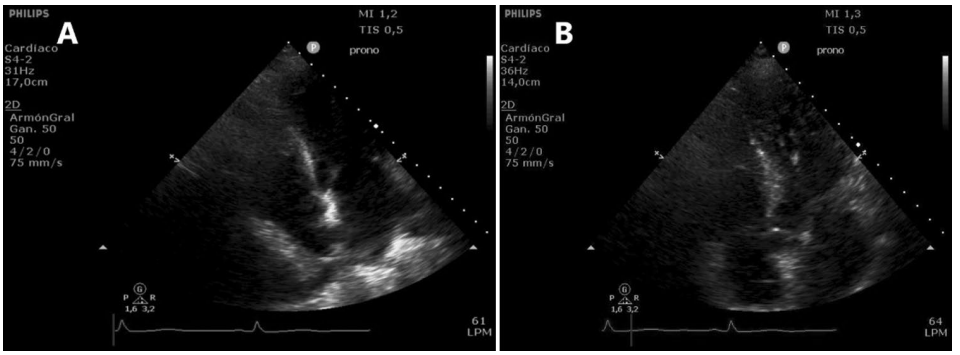


Fig. 2 Image in B mode in prone position. **A** Apical four chamber view. **B** Apical five chamber view

used continuous Doppler (CW) to estimate the pulmonary artery systolic pressure (PASP) using Bernoulli's formula. In the apical five chamber view, we systematically measured the velocity time integral (VTI) of the left ventricle outflow tract (LVOT) if its flow was correctly aligned (Figs. 3, 4).

Data collection

We collected the following variables from the electronic medical records: age, gender, comorbidities, diagnosis at admission, APACHE II score, respiratory parameters, vasopressor support, and in-hospital mortality.

The LV and RV basal diameter, LV systolic and diastolic function, the LVOT VTI, RV systolic function, and presence of valvular disease were obtained from the recorded images of ultrasonographic studies.

Statistical analysis

The data was analyzed with MedCalc® Statistical Software version 20.011 (MedCalc Software Ltd, Ostend, Belgium; https://www.medcalc.org; 2021). The frequencies and distribution of qualitative variables were expressed in absolute quantity and percentage, while quantitative variables were expressed as median and interquartile range. An exploratory analysis comparing groups of categorical variables was done with Chi-square test. A p-value less than 0.05 was considered statistically significant for all comparisons.

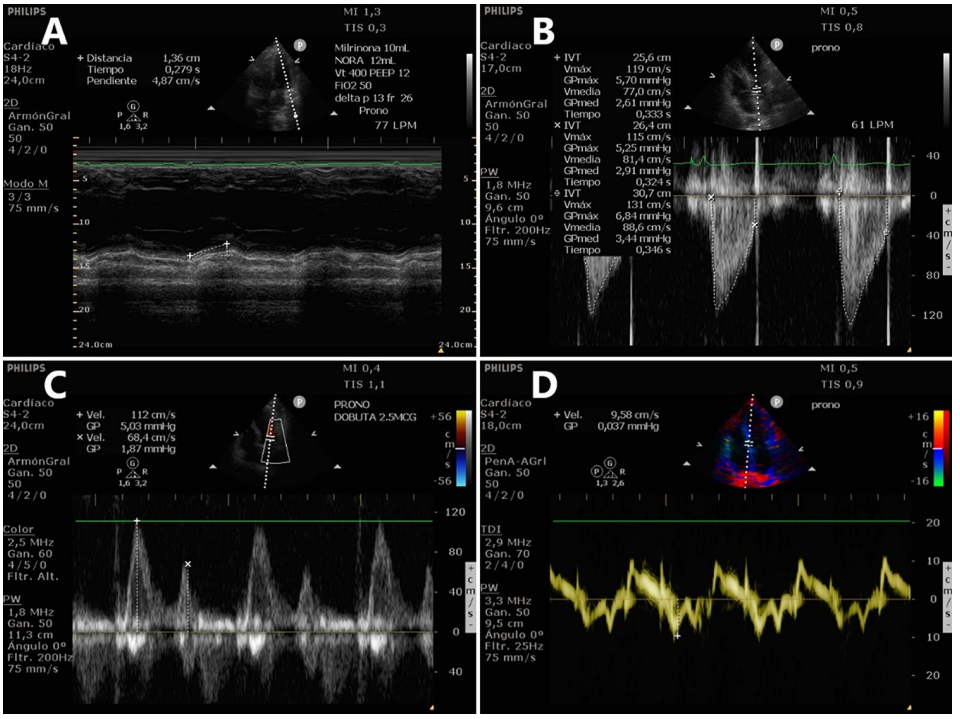


Fig. 3 Left ventricle assesment in prone position. **A** M mode measurement of lateral MAPSE in an apical four chamber view **B** PW at the LVOT above the aortic valve in the apical five chamber view with its VTI measurement. **C** PW at the transmitral flow in the apical four chamber view measuring of E and A waves velocities. **D** TDI measuring e' wave velocity at the septal border of the mitral ring in an apical four chamber view

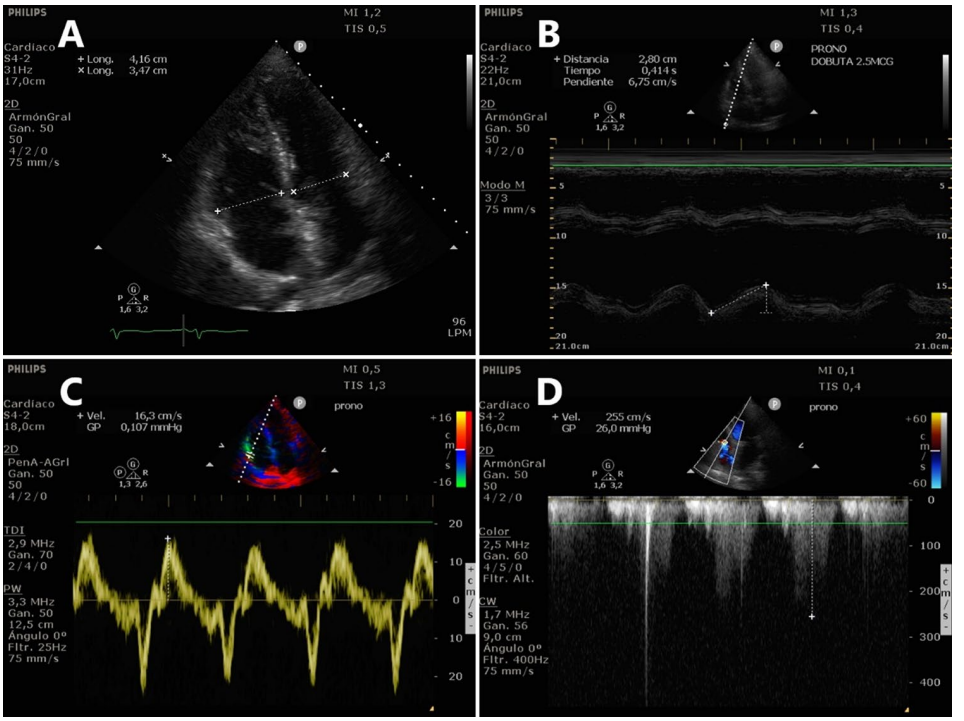


Fig. 4 Right ventricle assesment in prone position. **A** Measurement of the right and left ventricles basal diameter in B mode in an apical four chamber view. **B** M mode measurement of TAPSE in an apical four chamber view. **C** S′ wave velocity at the tricuspid ring in an apical four chamber view by TDI **D** CW at the tricuspid regurgitation flow measuring the peak velocity in an apical four chamber view

Results

From March 2020 to August 2021, a total of 93 ARDS patients underwent PPV. Of these, 35 patients (38%) were evaluated by TTE (Table 1). The median age was 56 (47–69) years, with a clear predominance of male subjects (80%). The most frequent comorbidities were arterial hypertension, obesity, dyslipidemia, smoking and diabetes. The cause of ARDS was COVID-19 pneumonia (C-ARDS) in the vast majority of cases (91.4%). The in-hospital mortality was 51.4%, with a median APACHE II score of 16 (12–20).

Most of the patients had suitable views for echocardiographic assessment. Overall, in 94.3% of the patients, it was possible to obtain an apical four chambers view and in 68.5% of the examinations, the apical five chambers view was satisfactory enough for LVOT VTI quantification.

The LV and RV systolic function was qualitatively assessed in the vast majority of the patients. Regarding the quantitative assessment, the septal MAPSE was measured more frequently (88.5%) than the lateral MAPSE (57.1%). The TAPSE was obtained in 91.4% of the patients. In most of the patients, we were able to measure the LVOT VTI and the RV/LV diameter. We assessed the diastolic LV function through the E and A wave ratio in more than half of the study population. Only 14.2% of the patients presented mild tricuspid regurgitation flow by color Doppler that allowed PASP estimation. Given the potential impact of obesity in the echocardiographic quality image, we performed an exploratory analysis of the feasibility of obtaining the echocardiographic variables in obese patients compared to non-obese patients with no statistical significance. Table 2 describes the number of measurements of echocardiographic variables and its comparison between obese and non-obese patients.

In our study, 30 of the patients had a preserved LV systolic function, while two had moderate to severe impairment. The median LVOT VTI was 20.6 (18.3–24.2) cm. All evaluated patients (94.3%) had a preserved RV systolic function ("eyeball"). The median value of RV/LV ratio was 0.8 (0.7–0.9). The TAPSE was 2.0 (1.8–2.2) cm and the tricuspid lateral ring S' wave was 11.6 (10.6–13.6) cm/sec. In none of the cases pericardial effusion or significant valvulopathies were found. The characteristics of echocardiographic variables are described in Table 3.

Discussion

The rate of PPV in ARDS was low for years, in spite of its benefits on gas exchange and mortality. In recent years, it has been steeply offered to mechanically ventilated patients with C-ARDS. Prone ventilation similarly improves oxygenation in C-ARDS and non-C-ARDS patients, but mortality remains high (45%) in both [6, 16]

The critical condition of these patients requires careful hemodynamic monitoring. In an international survey, more than half of the COVID-19 ICU patients (54%) had an echocardiographic examination performed, mostly with a transthoracic approach (97%) [17].

Table 1 Demographic and clinical characteristics of patients (n = 35)

Patient characteristics	
Age (years)[a]	56 (47–69)
Male, n (%)	28 (80)
Comorbidities, n (%)	
Arterial hypertension	23 (65.7)
Ischemic heart disease	4 (11.4)
Smoker	5 (14.2)
Asthma	2 (5.7)
Obesity	14 (40.0)
Dyslipidemia	6 (17.1)
Diabetes	5 (14.2)
Cancer	3 (8.5)
Immunosuppression	2 (5.7)
Diagnosis, n (%)	
COVID-19 Pneumonia	32 (91.4)
Bacterial pneumonia	2 (5.7)
Hemophagocytic syndrome	1 (2.9)
APACHE II Score[a]	16 (12–20)
PaO_2/FiO_2[a]	166 (124–235)
Tidal volume (ml)[a]	420 (393–458)
Tidal volume (ml/kg of PBW)	6.2 (5.9–6.8)
Plateau pressure, (cmH_2O)[a]	24 (22–27)
PEEP (cmH_2O)[a]	12 (10–15)
FiO_2 (%)[a]	50 (46–60)
Vasopressors, n (%)	14 (40)
In-hospital mortality, n (%)	18 (51.4)

PBW predicted body weight

[a]Median (IQR)

Due to the interposition of the air-filled lungs between the heart and the thoracic wall, mechanical ventilated patients proposed a challenge for TTE assessment, especially obese and prone patients. In those circumstances, a TEE assessment is recommended given the improvement in quality image [10]. Nevertheless, its implementation has inherent difficulties that were accentuated by the COVID-19 pandemic: the probe is not available in many of the ICUs, a TEE trained physician is needed, has higher risk of aerosol generation, and require high workload including proper equipment sterilization and longer exposure to the patient's infected particles [9, 11, 18].

As we do not have at disposal a TEE probe, we have been performing TTE assessment in PPV patients since 2015. Its practice and implementation increased considerably since 2020 with C-ARDS.

The objective of the study was not to measure the largest number of echocardiographic variables but to adopt a pragmatic and point of care approach in accordance with the catastrophic health situation with scarce resources. We had in consideration to limit the exposure time of the sonographer and to not create excessive burden given the staff shortage, as posed at the beginning of the pandemic [11, 19].

In our study, a transthoracic echocardiographic assessment was feasible in 94.3% of the evaluated patients. In them, we obtained a qualitative LV and RV systolic function assessment and in most of the examinations, we measured the LV/RV ratio. These findings are similar to the ones previously reported [12–14].

Table 2 Number of measurements and its comparison between obese and non-obese patients

Echocardiographic measurements n (%)				
LV systolic function	All n = 35	Obese n = 14	Non-obese = 21	P value
LV systolic function (eyeball)	33 (94.3)	14 (100)	19 (90.5)	0.241
LV basal diameter	27 (77.1)	12 (85.7)	15 (71.4)	0.33
Lateral MAPSE	20 (57.1)	10 (71.4)	10 (47.6)	0.169
Septal MAPSE	31 (88.5)	13 (92.9)	18 (85.7)	0.517
LVOT VTI	24 (68.5)	9 (64.3)	15 (71.4)	0.661
LV diastolic function				
Transmitral E and A wave	19 (54.2)	7 (50)	12 (57.1)	0.683
Lateral mitral annulus e′ wave	11 (31.4)	4 (28.6)	7 (33.3)	0.772
Septal mitral annulus e′ wave	12 (34.2)	4 (28.6)	8 (38.1)	0.567
E/e′ ratio	12 (34.2)	4 (28.6)	8 (38.1)	0.567
RV systolic function				
RV systolic function (eyeball)	33 (94.3)	14 (100)	19 (90.5)	0.241
RV basal diameter	33 (94.3)	14 (100)	19 (90.5)	0.241
TAPSE	32 (91.4)	14 (100)	18 (85.7)	0.144
Tricuspid annulus S′ wave	8 (22.8)	2 (14.3)	6 (28.6)	0.33
Other measurements				
PASP	5 (14.2)	1 (7.1)	4 (19)	0.33
RV/LV ratio	27 (77.1)	12 (85.7)	15 (71.4)	0.33

Table 3 Characteristics of echocardiographic variables

Echocardiographic variable	n = 35
LV systolic function	
Preserved LV systolic function, n (%)[a]	30 (85.7)
Mildly impaired LV systolic function, n (%)[a]	1 (2.8)
Moderately impaired LV systolic function, n (%)[a]	1 (2.8)
Severely impaired LV systolic function, n (%)[a]	1 (2.8)
LV basal diameter (cm)[b]	4.5 (4–5)
Lateral MAPSE (cm)[b]	1.2 (1–1.6)
Septal MAPSE (cm)[b]	1 (1–1.2)
LVOT VTI (cm)[b]	20.6 (18.3–24.2)
LV diastolic function	
E wave (cm/sec)[b]	86.3 (58.2–95.4)
A wave (cm/sec)[b]	70 (52.5–111.8)
Lateral mitral annulus e′ wave (cm/sec)[b]	13.3 (8.1–16.7)
Septal mitral annulus e′ wave (cm/sec)[b]	8.7 (6.4–9.8)
E/e′ ratio[b]	8.6 (2.7–9.6)
RV systolic function	
Preserved RV systolic function, n (%)	33 (94.3)
RV basal diameter (cm)[b]	3.7 (3.2–4)
TAPSE (cm)[b]	2.0 (1.8–2.2)
Tricuspid annulus S′ wave (cm/sec)[b]	11.6 (10.6–13.6)
Other measurements	
RV/LV ratio[b]	0.8 (0.7–0.9)
PSAP (mmHg)[b]	20.9 (9.7–25.1)

[a]Eyeball

[b]Median (IQR)

The LVOT VTI was quantified in 68.5% of the patients, limited due to impediment obtaining an apical five-chamber view. The quantitative LV systolic assessment by MAPSE was slightly lower than previous studies, mainly regarding the lateral MAPSE (57.1%). This may be explained because in some of our examinations we experienced difficulties obtaining a proper view of the LV's lateral free wall. Unlike some authors, we did not implement any specific maneuver to enhance our echocardiographic views. Giustiniano et al. deflated the lower thoracic sector of the air-mattress to place the probe and Ugalde et al. elevated the left shoulder with a pillow to create a space to fit in the transducer [14–20].

Considering that the obese population often present a challenge to obtain a good quality echocardiographic image, even as outpatients [21, 22], we performed an exploratory analysis comparing our echocardiographic measurements among obese and non-obese patients. There was no difference of statistical significance in this small series of cases.

The evaluation of the RV focused on the visual estimation of its systolic function, the measurement of the RV/LV basal diameter ratio, TAPSE in most cases, and the S′ wave in some, acknowledging that there is active research on the best assessment of this topic [23–27].

Our study shows that transthoracic echocardiography in prone position ventilation is feasible. It is a non-invasive, reproducible, and readily-available method that could be the preferred first tier assessment of the PPV patients.

We acknowledge several limitations in our research, principally due to its retrospective design. We were not able to control confounders or compare measurements. We performed TTE studies in only 38% of the prone ventilated patients during the study period, so a selection bias cannot be ruled out. We could not applied to the PRICES recommendation for reporting CCE data [28]; rather than having a protocol for exhaustive quantitative assessments, we used a goal-oriented TTE examination. As described in a recent systematic review of the literature, this circumstance was not unusual while depicting echocardiographic findings from COVID-19 patients [29]. Moreover, that study found a heterogeneous approach to data reporting, with the larger study included providing mainly qualitative information [30].

Nevertheless, we think this limited evaluation reflects real life care in several ICUs due to excessive workload and staff shortage.

Conclusion

Transthoracic echocardiography assessment could be used, even in suboptimal conditions, as an initial hemodynamic assessment of mechanical ventilated patients in prone position. In most cases, it allows to evaluate the LV and RV systolic function and the measurement of LVOT VTI as a surrogate of the SV.

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Author contributions Conceptualization: IC; Methodology: IC, VOC, RAG; Formal analysis and investigation: IC, VOC, RAG; Writing—original draft preparation: IC; Writing—review and editing: IC, VOC, RAG; Supervision: RAG, PMM, FMT.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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ORIGINAL PAPER

A modified subcostal view: a novel method for measuring the LVOT VTI

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Abstract

Purpose The velocity time integral (VTI) of the left ventricular outflow tract (LVOT) obtained in the apical view by echocardiography can be regarded as a surrogate for the stroke volume. In critically ill patients it is often difficult to obtain an appropriate apical view to assess the VTI. The subcostal view is more accessible, but while it allows a qualitative assessment of the heart, is not adequate for estimating a reliable LVOT VTI, given the inappropriate angle between the Doppler signal and the flow through the LVOT. We present a new modified subcostal view that allows a proper LVOT VTI measurement.

Methods This is a single-centre experimental, retrospective, and observational study using data from patients in a tertiary-care centre. We included adult patients admitted to the intensive care unit in the period from June 2020 to January 2022, who were evaluated by echocardiography and whose LVOT VTI was measured aligned with the Doppler signal in both the apical five-chamber view and the modified subcostal view.

Results A total of 30 patients were evaluated in the study period by ultrasonography. The Bland–Altman method analysis of the LVOT VTI measured in the apical view compared with that obtained in the subcostal view showed a bias of 0.8 (95% CI 0.39–1.21) with a 95% limit of agreement between − 1.35 (95% CI − 2.06 to − 0.64) and 2.96 (95% CI 2.25–3.67). The percentage error was calculated to be 23%. The Pearson correlation coefficient for the two forms of measurements showed an *R* value of 0.98 (95% CI 0.96–0.99).

Conclusion The LVOT VTI measured in a modified subcostal view is useful for estimating the value of the LVOT VTI obtained in an apical view.

Keywords POCUS · Echocardiography · Cardiac output · Subcostal view · Critical care

Introduction

In recent years, echocardiographic assessment has become a fundamental tool for monitoring the hemodynamics of critically ill patients. It enables calculating a patient's stroke volume (SV) and cardiac output (CO) based on the measurement of the left ventricle outflow tract (LVOT) velocity–time integral (VTI) in an apical five-chamber view and the LVOT area. The VTI represents the column of blood that travels through the LVOT during the systole [1, 2]. The SV and CO can be obtained by multiplying the LVOT VTI by the LVOT area. The latter is obtained by measuring the LVOT diameter in a parasternal long-axis view [2]. Given that the LVOT is constant in the same patient, it is possible to simplify the hemodynamic assessment and regard the LVOT VTI as a surrogate for the SV [3–7]. In critically ill patients it is often difficult to obtain an appropriate five-chamber view, due to the patient's position, mechanical ventilation, or the presence of thoracic drainages. The subcostal window is more accessible and allows a qualitative assessment of the heart. However, in this view, the alignment between the Doppler signal and the direction of the LVOT flow is not appropriate for estimating a reliable VTI [8]. We present a modified subcostal view that allows the LVOT VTI to be obtained.

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Material and methods

We present an experimental, observational, retrospective, and single-centre study, performed in our intensive care unit (ICU) between June 2020 and January 2022. The study was conducted in accordance with the principles of the Helsinki declaration and in compliance with the necessary precautions established by ethical and legal standards, preserving the identity of patients. The ethics and research committee of our institution approved this study and waived the requirement for informed consent. We included all adult patients admitted to the ICU who were evaluated by echocardiography and whose LVOT VTI was measured with the correct alignment with the Doppler signal, both in the apical five-chamber view and in the modified subcostal view.

Sample size

We used MedCalc® Statistical Software version 20.011 (MedCalc Software Ltd, Ostend, Belgium; https://www.medcalc.org; 2021) to calculate the sample size for the Bland–Altman plot, assuming a power of 0.8, an alpha value of 0.05, an expected mean difference of 1.1, an expected standard deviation of 0.95, and a maximum allowed difference between methods of 4, which resulted in a minimum required number of pairs of 24.

Ultrasonography

For ultrasonographic evaluation, we used a Philips Envisor HD or Sparq ultrasound machine (Philips Healthcare, WA, USA) with a phased array transducer. The modified subcostal view was obtained by placing the transducer below the subxiphoid region with an angle of approximately 45° from the skin, with its mark pointing upwards with the patient in supine and semi-recumbent position. We obtained an axial view of the aortic valve, with the right ventricle and pulmonary artery. From this position, the transducer was rotated slightly counterclockwise until the LVOT was visualized together with the aortic valve. Once the signal was aligned to the direction of the LVOT flow, the pulsed-wave Doppler was activated, and we positioned the sample volume just above de aortic valve to obtain the respective spectra (Figs. 1, 2 and 3). Subsequently, the LVOT VTI was obtained through the apical five-chamber window, considered to be the reference method, thereby ensuring a correct alignment between the Doppler signal and the flow direction. Three consecutive measurements of each method were made, and the recorded value corresponded to the average of the three.

Data collection

The data were collected from the patients' electronic medical records, as well as from stored ultrasonographic studies. The following variables were considered: age, gender, comorbidities, APACHE II score, cause for admission, requirement of mechanical ventilation, vasopressor and inotropic support, systolic function of the left ventricle (LV) and right ventricle (RV), presence of valvulopathies, apical LVOT VTI, and subcostal LVOT VTI.

Statistical analysis

The frequencies and distribution of the qualitative variables were expressed in absolute quantities and percentages, while the quantitative variables were expressed as means ± standard deviations and interquartile ranges according to the observed characteristics of the distribution. The normal distribution of the quantitative variables was evaluated with the Shapiro–Wilk test. To study the agreement between the LVOT VTI measurements obtained in the apical five-chamber view and the modified subcostal view, the Bland–Altman method was used. We calculated the percentage error derived by dividing the limits of agreement by the mean value of the measurements obtained with the established method [9]. The Pearson correlation coefficient was used to find an association between the variables. For all measurements, a 95% confidence interval (95% CI) showing no overlap was considered to be statistically significant.

Results

A total of 30 patients were evaluated in the study period by ultrasonography. Trauma was the most frequent reason for hospitalization (20%), followed by pneumonia (16.6%), septic shock (10%), and stroke (10%). The mean value of the APACHE II score was 15.7 ± 9.4. Of all patients, 56.6% required mechanical ventilation, 16.6% vasopressor support with noradrenaline, and 3.3% inotropic support with dobutamine. Based on the echocardiographic assessment, the majority of the patients had a normal LV systolic function (90%) and RV systolic function (96.7%). The mean value of the apical LVOT VTI measured in the standard view was 19.1 ± 6 cm, and the one obtained in the subcostal view was 18.2 ± 5.8 cm. Table 1 shows the demographic, clinical, and echocardiographic characteristics of the patients.

Table 1 Characteristics of the patients

Variable	$n = 30$
Age (years), median (SD)	51.4 (19.1)
Gender (male), *n* (%)	25 (83.3%)
Comorbidities	
Arterial Hypertension	9 (30%)
Diabetes	4 (13.3%)
Ischemic heart disease	4 (13.3%)
Dyslipidemia	1 (3.3%)
Obesity	1 (3.3%)
Cause for admission, *n* (%)	
Trauma	6 (20%)
Pneumonia	5 (16.6%)
Septic shock	3 (10%)
Stroke	3 (10%)
Gastrointestinal bleeding	2 (6.7%)
Other causes	11 (36.7%)
APACHE II score (points), mean (SD)	15.7 (9.4)
Mechanical ventilation, *n* (%)	17 (56.6%)
Noradrenaline vasopressor support, *n* (%)	5 (16.6%)
Dobutamine inotropic support, *n* (%)	1 (3.3%)
Left ventricle systolic function, *n* (%)	
Normal	27 (90%)
Moderately impaired	1 (3.3%)
Severely impaired	2 (6.7%)
Right ventricle systolic function, *n* (%)	
Normal	29 (96.7%)
Impaired	1 (3.3%)
Valvulopathies, *n* (%)	
Mild mitral regurgitation	2 (6.6%)
Moderate mitral regurgitation	1 (3.3%)
Mild tricuspid regurgitation	3 (10%)
Moderate tricuspid regurgitation	1 (3.3%)
Mild aortic regurgitation	5 (16.6%)
Apical LVOT VTI (cm), mean (SD)	19.1 (6)
Subcostal LVOT VTI (cm), mean (SD)	18.2 (5.8)

The Bland–Altman method analysis of the LVOT VTI value measured in the apical view compared with that obtained in the subcostal view showed a bias of 0.8 (95% CI 0.39–1.21) with a 95% limit of agreement between − 1.35 (95% CI − 2.06 to − 0.64) and 2.96 (95% CI 2.25–3.67). The analysis showed that 100% of the differences fell between the limits of agreement (Fig. 4). The percentage error was calculated to be 23%. The Pearson correlation coefficient for the two forms of measurements showed an *R* value of 0.98 (95% CI 0.96–0.99) (Figs. 5).



Fig. 1 Transducer position for the modified subcostal windows to visualize LVOT

Discussion

The cardiac output measurement is extremely useful in the management of critically ill patients. Echocardiographic assessment is the recommended non-invasive approach, especially for patients with shock [10]. The SV can be obtained by multiplying the LVOT area by the LVOT VTI from the apical five-chamber view. Considering that the diameter of the LVOT does not change in the same patient, the LVOT VTI can be regarded as a surrogate for the SV. However, it is not always possible to obtain an apical five-chamber view in critically ill patients. In this case, the subcostal windows could be an interesting alternative, allowing a qualitative hemodynamic assessment. Considering that in Doppler mode the velocity of a flow depends on the cosine of the angle formed between the Doppler signal's incidence and the direction of the flow, the alignment in this view is not adequate, so it is not recommended to measure the LVOT VTI in this view. Some authors have proposed using the angle correction function, but this option does not seem to be useful for correcting the velocity [8]. In this context, according to our knowledge, this is the first study that shows the possibility of measuring the LVOT VTI through a modified subcostal window, obtaining an adequate alignment between the Doppler signal and the LVOT flow.

Generally, when a new measurement technique of a variable is proposed, it is compared with the reference technique to obtain the degree of agreement between both methods and to validate the benefits of the new method [11]. Critchley and Critchley showed that the variation between the new method and the reference method of measuring cardiac output could be considered acceptable if the percentage error was less than 30% [12]. The percentage error calculated in our study was 23%, which would indicate that this novel method for estimating the

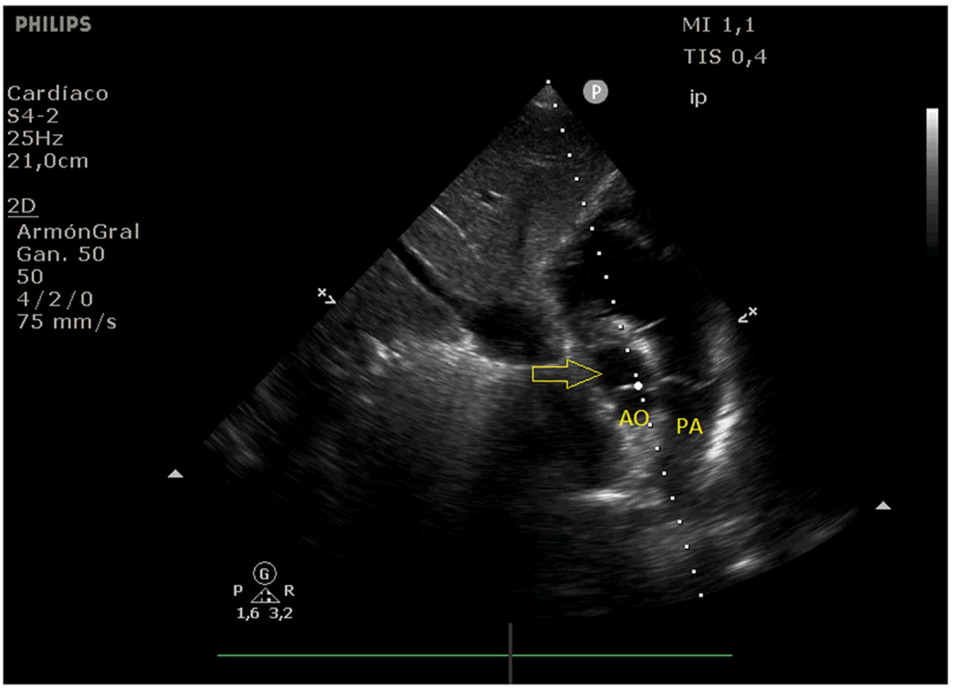


Fig. 2 B mode of the modified subcostal view that allows the visualization of the LVOT (yellow arrow) which is aligned with the dotted line representing the incidence of the Doppler signal (*AO* aorta, *PA* pulmonary artery)

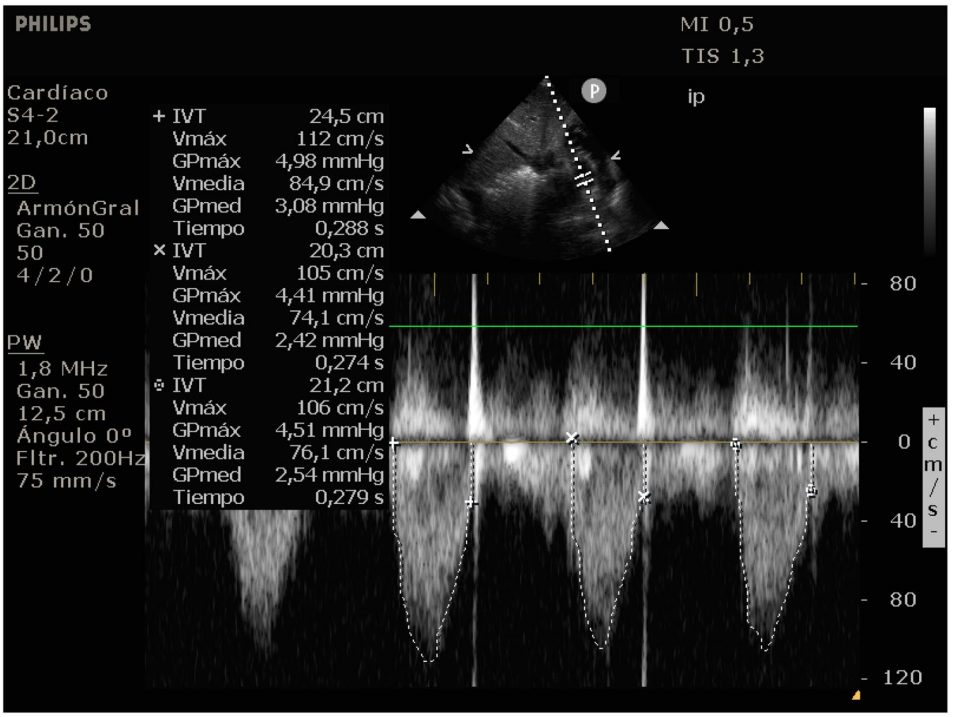


Fig. 3 Pulsed wave Doppler mode at the LVOT above the aortic valve in the modified subcostal view with its VTI measurement

LVOT VTI could be interchangeable with the conventional method and could be used when it is not possible to obtain an apical five-chamber view.

Our study has several limitations. First, we should mention the retrospective design, which reduces the control over multiple confounders. Second, only a single centre participated in the study. Third, interobserver reproducibility was not assessed, and finally, there was no blind operator.

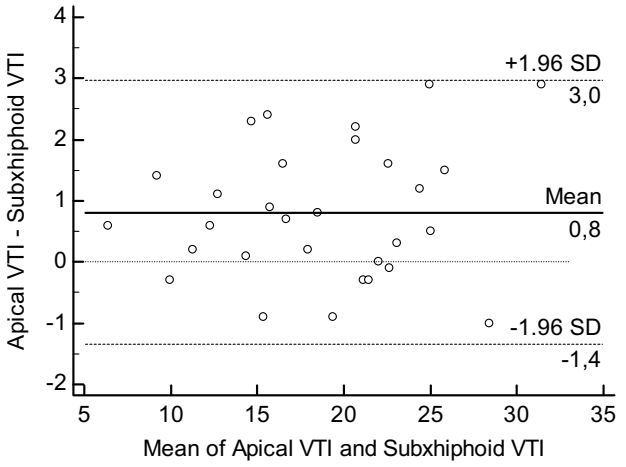


Fig. 4 Bland–Altman plot between LVOT VTI in apical and the modified subxiphoid views

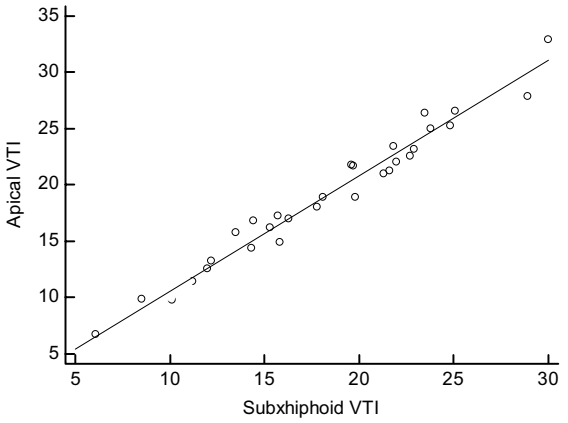


Fig. 5 Correlation between LVOT VTI in apical and the modified subxiphoid views

Conclusion

Even with the limitations described, our findings suggest that the LVOT VTI measured in the modified subcostal view, with the incidence of the Doppler signal aligned to the LVOT, is practically interchangeable with the LVOT VTI obtained in the conventional apical five-chamber view. It is, therefore, a useful alternative for assessing the LVOT VTI when the apical view is not available.

Authors contributions Conceptualization: Issac Cheong; Methodology: Issac Cheong; Formal analysis and investigation: Issac Cheong; Writing – original draft preparation: Issac Cheong; Writing – review and editing: Issac Cheong, Victoria Otero Castro; Supervision: Raúl A Gómez, Pablo M Merlo, Francisco M Tamagnone.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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CASE IMAGE

WILEY

Ultrasonography in the diagnosis of lung abscess

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Management of critically ill patients requires imaging techniques to optimize appropriate clinical decisions. Lung ultrasound has become an attractive new tool for assessing the lung in critically ill patients and could be easily performed at the bedside by the intensive care unit (ICU) physician.[1] We present a case of a patient with a lung abscess diagnosed by ultrasonography in the ICU.

A 36-year-old male with a clinical history of asthma was admitted to the emergency department with dyspnea and bronchospasm on physical examination. A chest X-ray was performed without any alterations. Due to respiratory insufficiency, he required orotracheal intubation and mechanical ventilation support, being admitted to the ICU. During the hospital stay, he presented persistent fever. Because in our ICU, we have extensive experience in point-of-care ultrasound, lung ultrasonography was performed as the initial diagnostic method. It showed a rounded hypoechoic image within a left basal consolidation, with the presence of power Doppler signal around, compatible with abscessed pneumonia (Figure 1). According to these findings, we directly requested a contrast-enhanced chest computed tomography without the need for a chest X-ray. It showed large nonaerated areas with subtotal involvement of the basal lung pyramids, with an image of intraparenchymal liquid density at the left basal pulmonary consolidation compatible with lung abscess (Figure 2). Fiberoptic bronchoscopy with bronchoalveolar lavage was performed, obtaining a purulent discharge, with a development in the culture of *Klebsiella pneumoniae*. The patient evolved favorably with antibiotic treatment extubating without complications, with subsequent discharge from ICU.

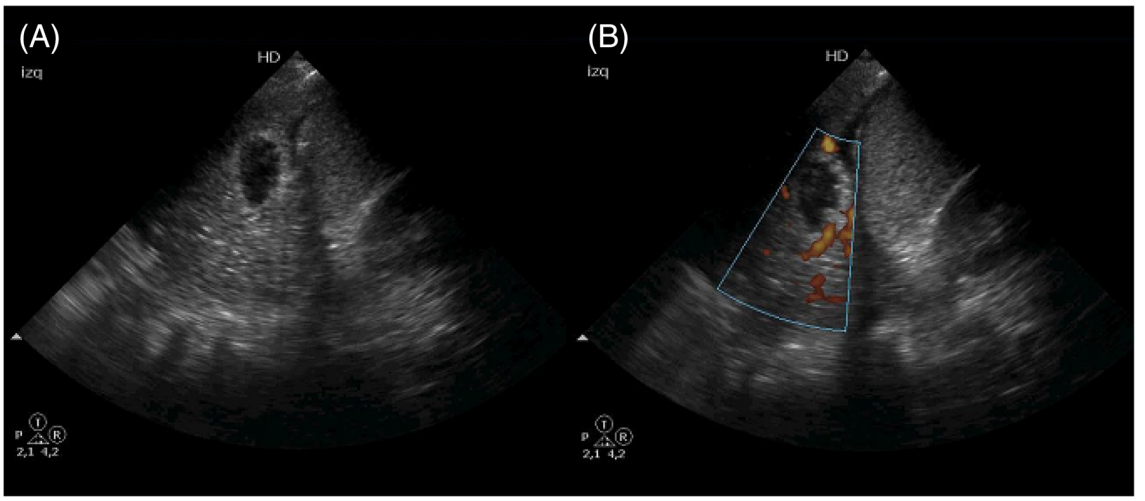


FIGURE 1 Lung ultrasound showing left basal consolidation with the presence of a rounded hypoechoic image within. (A) B-mode and (B) B-mode with power Doppler

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FIGURE 2 Contrast-enhanced chest computed tomography with an image of intraparenchymal liquid density at the left basal pulmonary consolidation compatible with lung abscess

Lung abscess is defined as a localized area of suppuration with the destruction of the lung parenchyma.[2] Despite the improvement of the prognosis of lung abscess with antimicrobial use, it remains an important respiratory disease with significant mortality.[3]

Traditionally, a chest X-ray is the initial diagnostic modality for lung abscess. However, diagnosis of a peripheral thoracic lesion containing an air–fluid level with this method can be difficult. Lung ultrasound is a simple and powerful imaging tool widely used in the survey of various peripheral lung disorders.[4]

In our case, the lung abscess was identified by ultrasonography as a hypoechoic image within a consolidation, with the presence of color Doppler signal around, as defined by the study of Chen et al. which showed a sensitivity of 94% and specificity of 100% for its diagnosis.[2]

This case is an example of the usefulness of lung ultrasound in the diagnosis of lung abscess in ICU.

CONFLICT OF INTEREST

The author declares that he has no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Echocardiography WILEY

CASE IMAGE

A new approach of non-invasive hemodynamic assessment by echocardiography in the intensive care unit: The right intercostal transhepatic window

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Abstract

Echocardiography has gained great acceptance in the management of critically ill patients, allowing non-invasive hemodynamic assessment, through conventional cardiac windows such as the parasternal, apical and subxiphoid views. However, it is not always possible to get good quality images because of the interposition of the lung or the presence of surgical drainages at the thoracic region. Based on these issues, we described a case in which echocardiographic assessment was possible through a novel approach: the right intercostal transhepatic window (RITW), through which we diagnosed significant aortic regurgitation in a patient with pulmonary edema.

KEYWORDS
echocardiography, hemodynamic assessment, POCUS, point-of-care ultrasound

Echocardiography has gained great acceptance in the management of critically ill patients, allowing non-invasive hemodynamic assessment, through conventional cardiac windows such as the parasternal, apical and subxiphoid views.[1] However, it is not always possible to get good quality images because of the interposition of the lung or the presence of surgical drainages at the thoracic region. Based on these issues, we described a case in which echocardiographic assessment was possible through a novel approach: the right intercostal transhepatic window (RITW).

A 76-year-old female patient with a history of hypertension, diabetes, and pancreatic tumor was admitted to the medical clinic service with a diagnosis of cholangitis. She underwent urgent percutaneous drainage of the bile duct. She developed respiratory failure and was brought to the intensive care unit (ICU) requiring orotracheal intubation and mechanical ventilation support upon her arrival.

A lung ultrasound was performed after she was stable and showed the presence of bilateral confluent B lines compatible with the alveolar-interstitial syndrome. Both basal lung regions were evaluated, and the presence of bilateral pleural effusion in moderate degree was noted. During her evaluation we realized that it was possible to see the heart cavities, using the hepatic parenchyma as an acoustic window by positioning the phased array transducer in the right anterior axillary line at the level of the 5° intercostal space with the transducer marker pointing upwards (Figure 1).

Through this unconventional view, it was possible to evaluate the preserved systolic function of both ventricles, as well as the absence of significant pericardial effusion. By tilting the transducer slightly towards the dorsal region, the left ventricle outflow tract (LVOT) and the aortic valve were easily evaluated. Because the LVOT was perfectly aligned with the Doppler signal, the pulsed wave Doppler (PWD) was used placing the sample volume just below the aortic valve, obtaining the spectrum of the LVOT, and we observed a flow spectrum compatible with aortic regurgitation (Figure 2). We confirmed this by color and continuous wave Doppler (CWD) at the aortic valve, which was compatible with moderate degree aortic regurgitation[2] (Figure 3). These findings were corroborated with the corresponding measurements and the E/et' ratio taken through the apical view which showed significant diastolic dysfunction with the final diagnosis of cardiogenic pulmonary

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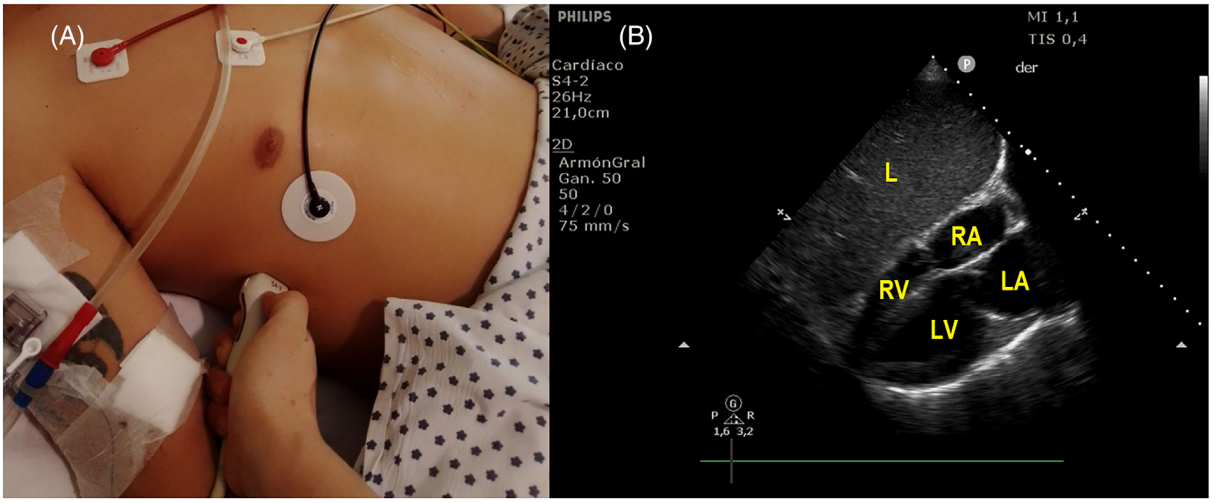


FIGURE 1 RITW. (A) Positioning of the transducer at the anterior axillary line to obtain the window. (B) Visualization of the heart chambers in B-mode (RA, Right atrium; LA, left atrium; RV, right ventricle; LV, left ventricle; L, liver)

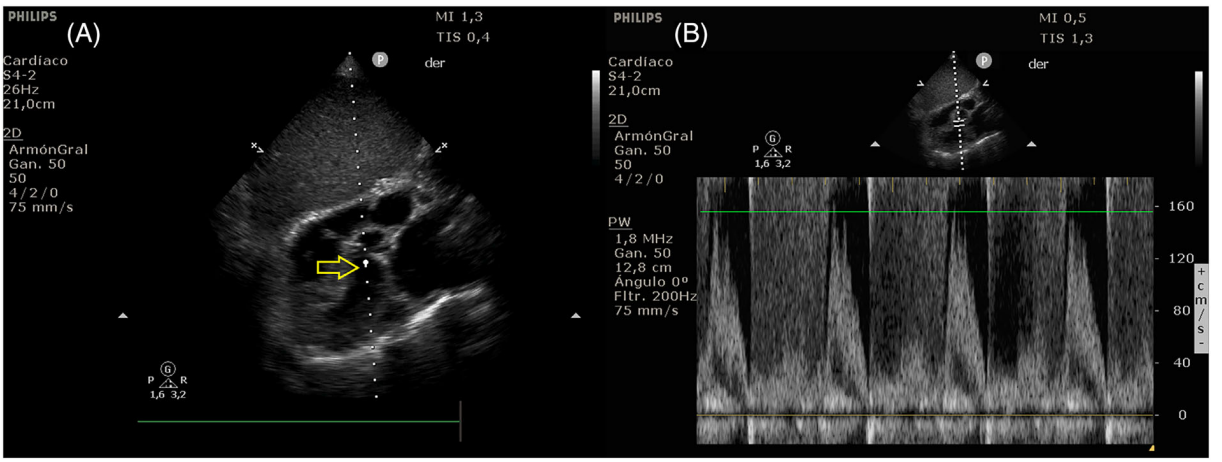


FIGURE 2 Five chambers at the RITW. (A) LVOT visualization in B mode (yellow arrow). (B) PWD at the LVOT, in which aortic regurgitation flow spectrum is visualized partially due to aliasing

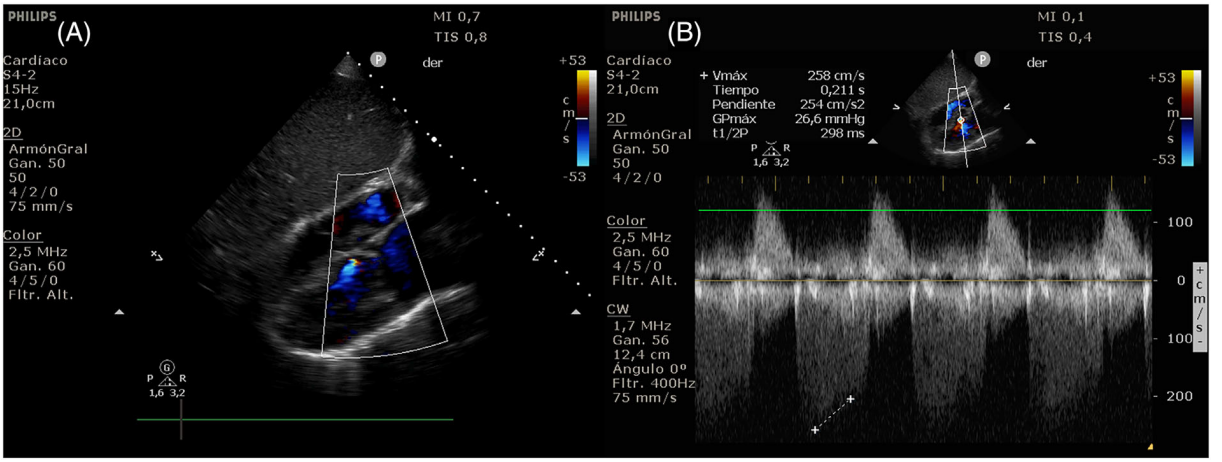


FIGURE 3 Assessment of aortic regurgitation through the RITW. (A) With color doppler. (B) With CWD

edema.[3] After this initial evaluation, the patient was put on Nitroglycerin and Furosemide to achieve negative fluid balance. After 24 h the oxygenation was improving and the echocardiographic parameters showed a decreasing E/et' ratio and aortic insufficiency. Also, B lines in the lung ultrasound evaluation were considerably less. The patient was extubated within three days of her arrival and was later discharged to the general ward.

To our knowledge, we have not found reports about this novel form of echocardiographic evaluation in the available medical literature. As a limitation, it is important to mention that it was not possible to evaluate diastolic function by this method since the tissue Doppler signal of this window was not parallel to the mitral annulus.

Despite this, our case shows the usefulness of the RITW as a new echocardiographic approach for overall assessment of systolic function and gross valvular assessment in critically ill patients.

CONFLICTS OF INTEREST

The author declares that he has no conflict of interest.

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CASE IMAGE

SONOGRAPHY WILEY

A case of heart point sign by lung ultrasound in a patient with pneumothorax

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1 | INTRODUCTION

Pneumothorax is a frequent problem in the intensive care unit. Lung ultrasound has demonstrated to be a useful method in its diagnosis with high sensitivity and specificity.[1,2] The pattern observed in pneumothorax are the abolition of lung sliding and the lung point sign defined as an inspiratory-synchronized change from lung sliding to no lung sliding in a specific location on the chest wall in B and M-mode.[2] We describe a case of pneumothorax diagnosed through a variant of the lung point sign at the left hemithorax: the heart point sign.[3]

2 | CASE PRESENTATION

A 29-year-old patient suffering polytrauma from a car-on-car collision was admitted to an Emergency Service of another Hospital with diagnosis of bilateral hemopneumothorax associated with costal fractures. He required orotracheal intubation, mechanical ventilation support and placement of pleural drainage tubes, being referred to the intensive care unit (ICU) of our Hospital with high oxygen requirements. An ultrasonographic evaluation of the lung was performed, showing the absence of lung sliding and the presence of the lung point sign in both hemithorax (Figure 1). When evaluating the heart at the short axis

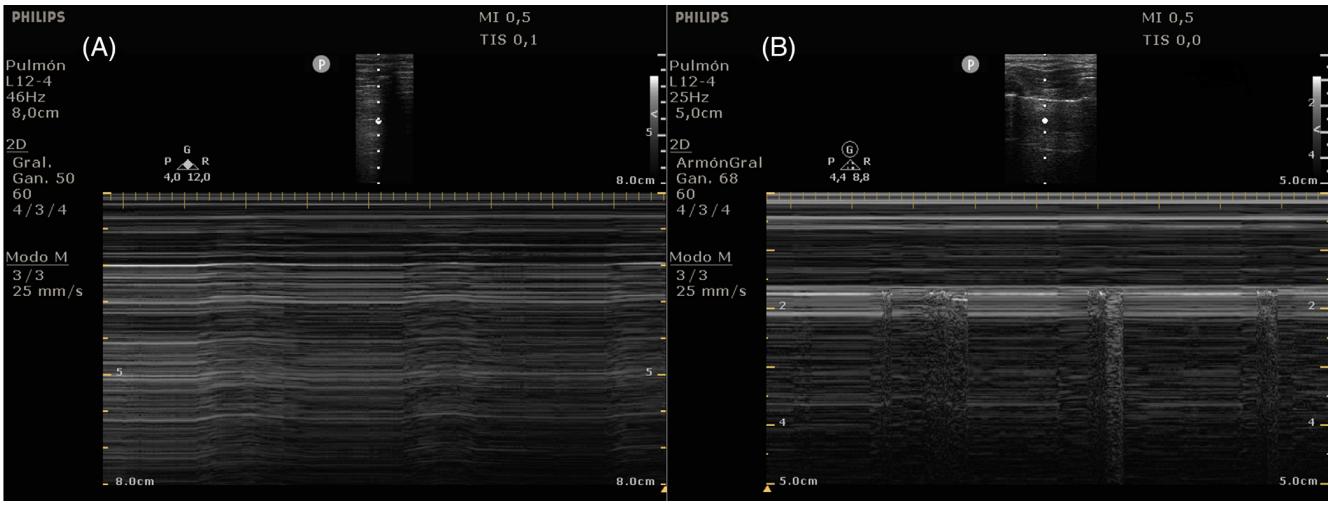


FIGURE 1 Pneumothorax by lung ultrasound in M mode. (A) Abolished lung sliding demonstrated by the stratosphere sign. (B) Lung point sign demonstrated by the alternance between the strastophere and seashore signs

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FIGURE 2 Chest CT showing bilateral pneumothorax

view, the visualization of the heart during diastole was evidenced disappearing suddenly during systole with the interposition of an image of absence of lung sliding with A lines which is known as the heart point sign[3] (Video 1). Computed tomography (CT) of the chest confirmed the findings (Figure 2). Thoracic surgery performed video-assisted thoracoscopy with pulmonary decortication, reduction of rib fractures, and replacement of bilateral pleural drainages. The patient improved his clinical condition being extubated in the short term and being discharged from the ICU.

The diagnosis of pneumothorax by ultrasound is based on the absence of the lung sliding and B lines, and the presence of A lines and the lung point sign.[1] The heart point sign was described by Stone et al, as a phenomenon specific to pneumothorax. During diastole, the heart increases in size as it fills with blood, which causes the displacement of air from the precardiac space, which can be visualized by ultrasound. On the other hand, during systole, when the heart empties, the air returns to interpose itself in the precardiac space, preventing its visualization by ultrasound.[3]

There are some pathologies that can be easily confused with pneumothorax, such as pneumomediastinum and pulmonary bullae, so its distinctive ultrasonographic characteristics must be known. In pneumomediastinum, transthoracic ultrasonography demonstrates poor visualization of the heart with A lines on the parasternal and apical views unrelated to the cardiac cycle, along with normal visibility on the subxiphoid view.[4] In pulmonary bullae, unlike pneumothorax, pulmonary sliding is present due to the absence of air between the parietal and visceral pleura.[5] This case demonstrate the usefulness of the heart point sign in the diagnosis of pneumothorax by lung ultrasound.

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CONFLICT OF INTEREST

The author declares that he has no conflict of interest.

INFORMED CONSENT

Written consent was obtained from the patient before the publication of this case.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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CASE VIDEO

SONOGRAPHY WILEY

A case of lung point sign in a patient with pneumothorax under high frequency oscillatory ventilation

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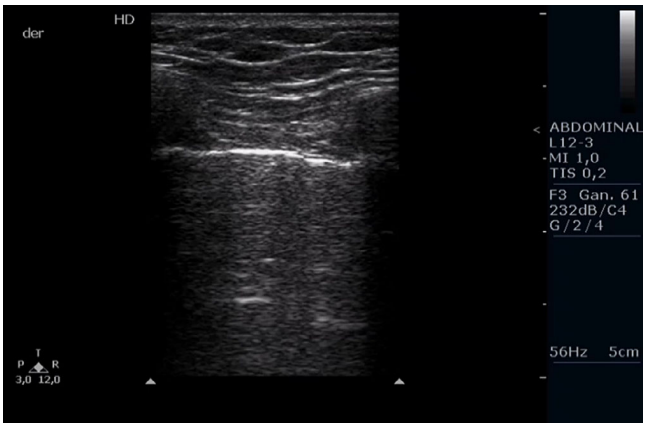
Pneumothorax is a frequent problem seen in the intensive care unit (ICU). Lung ultrasound has demonstrated to be a useful method in the diagnosis of pneumothorax with high accuracy. The normal lung is characterized by the presence of lung sliding (Video 1). The pattern observed in pneumothorax by ultrasound are the abolition of lung sliding (Video 2) and the lung point sign (Video 3) which has been shown to have 100% specificity for its diagnosis.[1] The lung point sign is defined as an respiratory-synchronized change from lung sliding to no lung sliding in a specific location on the chest wall.[2] We present a case of pneumothorax with the lung point sign in a patient under high frequency oscillatory ventilation (HFOV).

A 16-year-old patient suffering respiratory insufficiency due to acute respiratory distress syndrome was admitted to intensive care unit (ICU) requiring HFOV. Because he evolved with greater hypoxemia, we evaluated the lung with point-of-care ultrasound (POCUS). It showed at the anterior right chest the presence of the lung point sign compatible with pneumothorax. Specially, the lung sliding presented a high-frequency movement in accordance with its ventilator mode, which made the lung point sign so peculiar (Video 4). The diagnosis was confirmed by chest X-ray. The patient required the placement of a right pleural drainage tube, evolving with improved oxygenation.

It should be noted that there are situations where ultrasonography may fail to correctly rule a pneumothorax in or out. False-positive diagnosis of it has been described in bullous lesions and severe chronic obstructive pulmonary disease. Also, morbid obesity may present difficulty due to loss of resolution. In some cases, subcutaneous emphysema may mimic pleural motion and other artifacts similar to B-lines. In pleural adhesions, air may be trapped in different areas. In such cases, the lung point sign may be inaccurate to diagnose pneumothorax.[3]

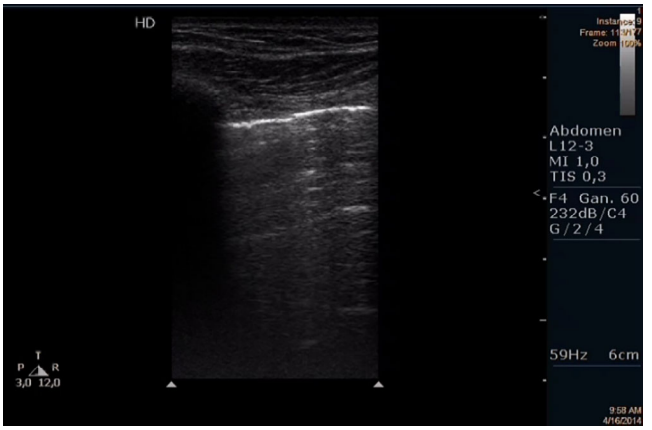


VIDEO 1 Lung sliding.
Video content can be viewed at https://onlinelibrary.wiley.com/doi/10.1002/sono.12331

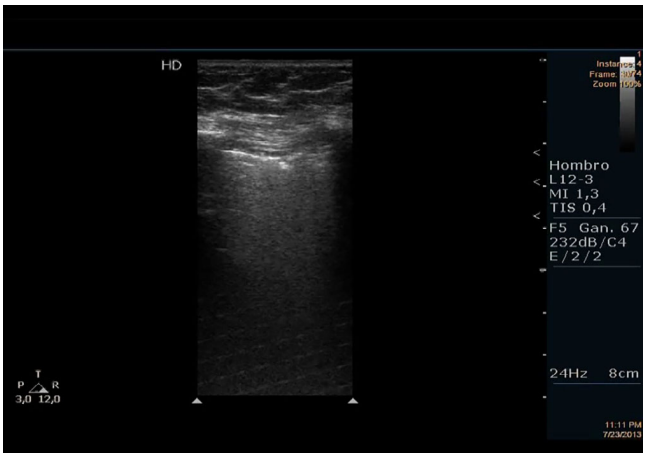


VIDEO 2 Lung sliding.
Video content can be viewed at https://onlinelibrary.wiley.com/doi/10.1002/sono.12331

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VIDEO 3 Lung sliding.
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VIDEO 4 Lung sliding.
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CONFLICT OF INTEREST

The author declares that he has no conflict of interest.

ETHICS STATEMENT

Written consent was obtained from the patient before the publication of this case.

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Journal of Ultrasound
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ORIGINAL PAPER

Usefulness of ultrasound in the diagnosis of nosocomial maxillary sinusitis in patients with severe Covid-19 pneumonia: a retrospective study

Issac Cheong[1,2] · Victoria Otero Castro[1] · Raúl Alejandro Gómez[1] · Francisco Marcelo Tamagnone[2]

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Abstract

Purpose Severe Covid-19 pneumonia frequently presents with infective complications as bacterial and fungal infections, nosocomial maxillary sinusitis is one of them. We describe the role of ultrasonography in the diagnosis of nosocomial maxillary sinusitis in patients undergoing mechanical ventilation due to severe Covid-19 pneumonia.

Methods Patients with severe pneumonia due to Covid-19 requiring mechanical ventilation and had maxillary sinus occupation by ultrasonography were retrospectively enrolled at intensive care unit of Sanatorio De los Arcos, Buenos Aires, Argentina, between March 2020 and May 2021.

Results We evaluated a total of 171 patients with severe Covid-19 pneumonia requiring mechanical ventilation during the study period. We included 26 patients that had maxillary sinus occupation by ultrasonography (15.2%). Out of the 26 patients, in 17 the diagnosis was confirmed by surgical drainage of the maxillary sinus with positive cultures with a positive predictive value of 85%. In 34.6% of the cases the maxillary sinus disease was unilateral and in 38.4% bilateral. A sinus computed tomography (CT) was performed in 30.7% of the patients. Most of the patients underwent to surgical drainage without a CT scan (66.7%).

Conclusions Ultrasonography is a useful tool for the detection of maxillary sinus infection in patients with severe Covid-19 pneumonia and avoids to perform a CT scan and, therefore, unnecessary transfers.

Keywords POCUS · Ultrasonography · Sonography · Maxillary sinusitis · Covid

Introduction

Secondary bacterial infections are important risk factors contributing to the severity and mortality of Covid-19 patients. In Wuhan was reported 15% of secondary infections, with higher incidence among non-survivors [1]. Patients with severe Covid-19 pneumonia requiring mechanical ventilation are more prone to develop nosocomial infections, as maxillary sinusitis [1–4]. The gold standard for maxillary sinus occupation is a sinus computed tomography (CT). The confirmed diagnosis of nosocomial maxillary sinus infection is bacterial growth in cultures or the presence of macroscopic pus during surgical drainage [3].

In order to perform a CT scan, the patient needs to be transfer outside the intensive care unit. Given that in-hospital transfers present frequent adverse effects (up to 71%) performing a bedside diagnosis may be safer for patient care and reduce workload [4]. Also when the healthcare system reaches saturations, as in Covid-19 pandemic, strategy must be adjusted to fairly allocate scarce resources, in order to maximize the benefit of the greatest number of patients [5].

Ultrasonography showed to be a useful tool for the diagnosis of maxillary occupation with adequate levels of sensitivity and specificity, in addition to its speed, safety, and reproducibility [6]. We describe our experience regarding the role of ultrasonography in the diagnosis of nosocomial maxillary sinusitis in patients undergoing mechanical ventilation due to Covid-19 pneumonia.

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Materials and methods

We conducted a retrospective, single-center observational study in the intensive care unit of Sanatorio De Los Arcos (Buenos Aires, Argentina) from March 2020 to May 2021. The study was conducted following the principles of the Helsinki Declaration and within the precautions established by ethical and legal standards, preserving the identity of patients and having obtained written informed consent of the participants for the use of their data.

We included all the adult patients diagnosed with Covid-19 pneumonia admitted to the intensive care unit with mechanical ventilation requirement with suspected infection and an image compatible with maxillary sinus occupation by ultrasonography.

Data collection

We collected the following variables from the electronic medical record: age, sex, comorbidities, APACHE II score, shock, days of mechanical ventilation, mortality, partial or total sinus occupation, unilateral or bilateral affection, fever, leukocytosis, CT scan sinus infection confirmation, surgical drainage requirement, and antibiotic treatment. We reviewed the stored ultrasonographic studies.

Ultrasonographic assessment

For the ultrasonographic evaluation, we used a Philips Envisor HD or Sparq ultrasound machine (Philips Healthcare, MA, USA) with a phased array S2–4 transducer. We placed the probe in each upper maxillary bone pointing toward right. In normal conditions, the presence of air causes reverberation artifices such as parallel echoes. When the maxillary sinus is fully occupied, we can observe its content (anechoic liquid) surrounded by hyperechoic sinus walls. Partial identification of the posterior wall is compatible with partial sinus occupation [2, 7] (Figs. 1, 2).

Protocol of study

We performed an ultrasonographic evaluation of both maxillary sinuses in all patients with suspected maxillary sinusitis as a probable nosocomial infection source. In transferable patients, we requested a sinus CT scan. If the patient presented ultrasonographic sinus occupation, a surgical drainage was performed. The definitive diagnosis of maxillary sinusitis was confirmed when there was bacterial growth in cultures in the material obtained during surgical drainage.

Statistical analysis

The frequencies and distribution of the qualitative variables are expressed in absolute quantity and percentage, while

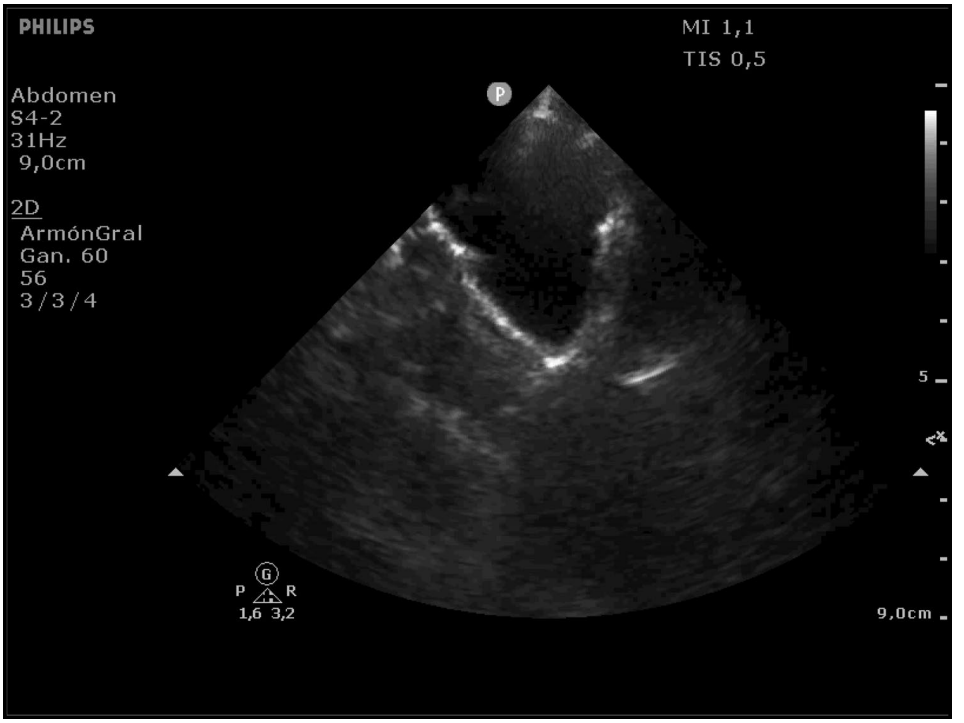


Fig. 1 Image in B mode using a phased array transducer placed in the upper maxillary bone showing total occupation of the maxillary sinus. The anechoic liquid content allows the display of the hyperechoic maxillary sinus walls

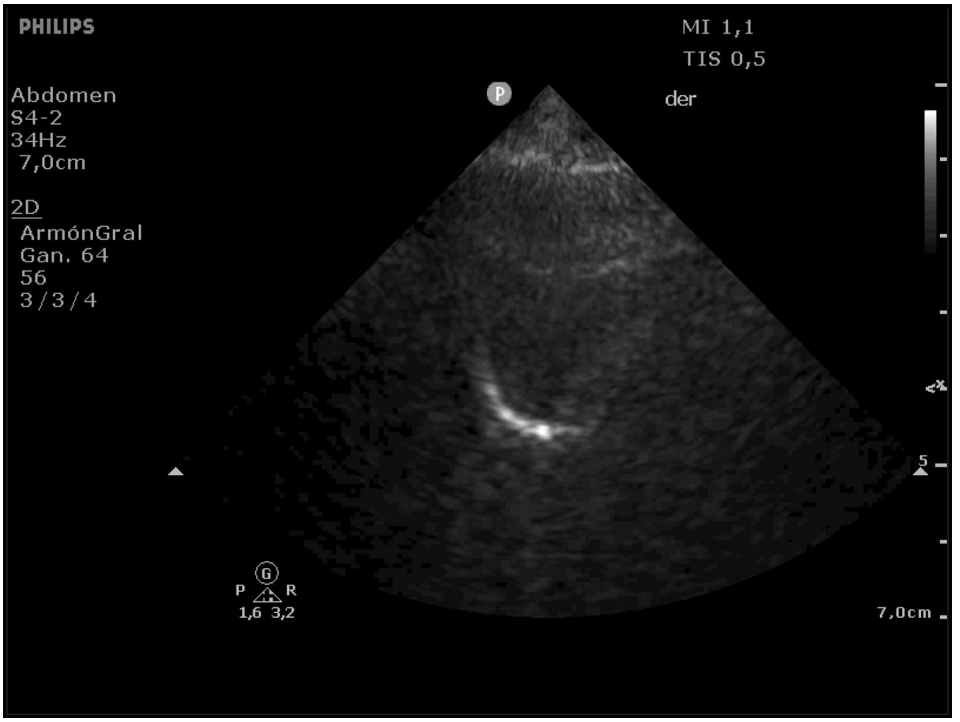


Fig. 2 Image in B mode using a phased array transducer placed in the upper maxillary bone showing partial occupation of the maxillary sinus. The anechoic liquid content only allows the display of the hyperechoic posterior portion of the maxillary sinus walls

the quantitative variables are presented as mean ± standard deviations, or median and interquartile range, according to the observed characteristics of the distribution. The normal distribution of quantitative variables was analyzed with the Shapiro–Wilks test. As exploratory analysis we calculated the positive predictive value of ultrasonography for the diagnosis of nosocomial maxillary sinusitis.

Results

A total of 171 patients with severe Covid-19 pneumonia requiring mechanical ventilation were evaluated during the study period, 26 of them had maxillary sinus occupation diagnosed by ultrasonography (15.2%). Table 1 describes the demographic characteristic of the patients.

Table 1 Demographic characteristics of patients included in the study

Variables	All ($N=26$)	Drained ($N=18$)	Not drained ($N=8$)
Age (IQR)	59 (49–69)	56 (49–72)	67 (54–70)
Male gender	21 (80.7%)	14 (66.7%)	3 (33.3%)
Comorbilities			
Hypertension	17 (65.3%)	12(70.6%)	5 (29.4%)
Dyslipemia	6 (23%)	5 (71.4%)	2 (28.6%)
Diabetes	6 (23%)	4 (66.7%)	2 (33.3%)
Obesity	8 (30.7%)	8 (100%)	0
Ischemic heart disease	3 (11.5%)	1 (33.3%)	2 (66.7%)
Atrial fibrillation	2 (7.6%)	0	2 (100%)
Ex-smoker	5 (19.2%)	4 (80%)	1 (20%)
Asthma	2 (7.6%)	2 (100)%	0
APACHE II score	14 ± 6	13.7 ± 6.6	14.6 ± 4.8
Shock	6 (23%)	4 (66.7%)	2 (33.3%)
Days on mechanical ventilation	24.1 ± 10.2	26.6 ± 9.6	21.2 ± 11.1
Mortality	7 (26.9%)	4 (57.1%)	3 (42.9%)

IQR interquartil range

Table 2 Clinical and ultrasonographic characteristics of the patients

Variables	All ($N=26$)	Drained ($N=18$)	Not drained ($N=8$)
Total and unilateral affection	9 (34.6%)	7 (26.9%)	2 (7.7%)
Partial and unilateral affection	7 (26.9%)	2 (7.7%)	5 (19.2%)
Bilateral affection	10 (38.4%)	9 (34.6%)	1 (3.8%)
Confirmatory sinus CT	8 (30.7%)	6 (23%)	2 (7.7%)
Cause of suspicion			
Fever	18 (69.2%)	13 (50%)	5 (19.2%)
Leukocytosis	11 (42.3%)	8 (30.7%)	3 (11.5%)
Antibiotic treatment	23 (88.4%)	18 (69.2%)	5 (19.2%)

Table 3 Ultrasonographic characteristics of maxillary sinus and diagnosis

Variables	Confirmed	Discarded
Total and unilateral affection	6 (85.7%)	1 (14.3%)
Partial and unilateral affection	2 (75%)	1 (25%)
Bilateral affection	9 (90%)	1 (10%)

In 38.4% of cases, the maxillary sinus occupation was total and unilateral, 26.9% partial and unilateral, and 38.4% bilateral. We performed a sinus CT in 30.7% of the patients. Most of the patients underwent to surgical drainage without a CT scan (66.7%).

Of the 26 patients, 18 patients had surgical drainage. Among them, 17 patients presented bacterial growth in the cultures. The surgical drainage was not performed in 8 patients. In the case of 1 patient, the care team decided to treat a different source of infection, 2 patients had coagulopathy and 3 patients were not within the goals of care; therefore these 6 patients were not included in the analysis for diagnostic accuracy. In the other 2 cases with partial occupation, sinusitis was ruled out by a CT scan with evidence of mucosal thickening. Among all surgical drainages, only 1 case of total affection of the maxillary sinus did not develop bacterial growth in culture. The positive predictive value of ultrasonography for the diagnosis of nosocomial maxillary sinusitis was 85% compared to the gold standard of bacterial growth in cultures in the material obtained during surgical drainage. Patients with partial and unilateral sinus occupation had a positive predictive value of 75%, patients with total and unilateral occupation 85,7%, and those with bilateral occupation 90%. Clinical and ultrasonographic characteristics are described in Tables 2 and 3. The most frequent germs involved were gram-negative bacilli followed by gram-positive cocci. Isolated microorganisms are described in more detail in Table 4.

Table 4 Microorganisms isolated from affected maxillary sinuses

Gram-negative bacilli	
Pseudomonas	3
Klebsiella	3
Escherichia coli	2
Enterobacter	2
Acinetobacter	1
Gram-positive cocci	
Streptococcus	5
Enterococcus	2
Staphylococcus aureus	2
Coagulase-negative Staphylococcus	1
Gram positive bacilli	
Corynebacterium	1
Lactobacillus	1
Fungi	
Candida	2
Anaerobes	
Actinomyces	1

Discussion

The sinus CT scan is the gold standard for the evaluation of maxillary sinus occupation. However, it requires to transfer critically ill patients, which has been shown to be associated with adverse effects and increased workload during the Covid-19 pandemic [4, 8, 9]. Ultrasonography is a useful tool for diagnosis and follow-up in the intensive care unit. It is safe, cost-effective, and can be performed at patient's bedside. Hilbert et al. demonstrated a sensitivity of 100% and specificity of 96.7% for the diagnosis of maxillary sinusitis in patients under mechanical ventilation [10]. Our study showed that CT scan was not essential for the confirmation of maxillary sinus occupation and consequently surgical drainage in 66.7% of cases. In our study, the ultrasonography has a high positive predictive value (85%) for the diagnosis of maxillary nosocomial sinusitis, mainly when the occupation of the maxillary sinus is total or bilateral.

In the context of a pandemic, being able to perform a bedside diagnosis of maxillary sinusitis, without exposing the patient to an in-hospital transfer, may be safer for patient care, lower work burden and cost-effective.

Our research has several limitations, mainly due to the retrospective design. We were not able to control confounders, not all patients with an image of maxillary sinus occupation by ultrasonography had a CT scan or were surgically drained, and it is a single center study. However, it is a novel method to assess maxillary sinusitis and all the ultrasonographic studies were performed by the same operator, limiting an interobserver variability.

Conclusion

According to our experience, ultrasonography is a useful tool for the diagnosis of nosocomial maxillary sinusitis in patients with severe Covid-19 pneumonia and may avoid unnecessary transfers in critically ill patients.

Author contributions Conceptualization: IC, RAG. Formal analysis: IC, VOC, RAG. Investigation: IC. Supervision: FMT. Writing—original draft: IC. Writing—review editing: IC, VOC, RAG.

Declarations

Conflict of interest The author declares that they have no conflict of interest.

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Utility of transoral and transcranial ultrasonography in the diagnosis of internal carotid dissection: a case report

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CASE REPORT

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Background: Internal carotid artery (ICA) dissection is one of the most common causes of stroke in young patients. Ultrasonographic assessment of the carotid artery is the method of choice for early detection. The use of the conventional technique for diagnosis is limited since dissection frequently occurs in the distal portion of the ICA.
Case Report: We describe the case of a 48-year-old Argentinian woman with malignant cerebral infarction secondary to ICA dissection. We diagnosed her with neck vessels using conventional ultrasonography and a transoral ultrasonographic approach combined with transcranial ultrasonography.
Conclusion: We recommend the transoral assessment of ICA combined with ultrasonography as a noninvasive method for the diagnosis of ICA dissection.

Keywords: Carotid artery, internal, dissection; Ultrasonography; Stroke

INTRODUCTION

Internal carotid artery (ICA) dissection is one of the most common causes of stroke in young patients. Approximately 70% of affected patients are under 50 years of age, and the annual incidence is 2.6–2.9 cases per 100,000 [1,2]. Ultrasonography (US) is the method of choice for the initial assessment of ICA dissection given its safety, rapid availability, and prompt bedside use. When dissection occurs in the distal portion of the ICA, it becomes difficult to detect, and it can be evaluated indirectly through its flow characteristics in Doppler mode. Transoral ultrasonographic evaluation of the ICA is a novel technique that allows the assessment of the distal portion of the vessel. The combination of transcranial Doppler US and evaluation of intracranial vessels can be useful for the diagnosis of ICA dissection. We present a case in which conventional and transoral ICA US was used in combination with transcranial Doppler to diagnose ICA dissection.

CASE REPORT

A 48 year-old Argentinian woman with no relevant clinical history presented to a peripheral hospital with right brachiocrural paralysis and central facial palsy. Prior to symptom onset, she underwent a neck massage for a headache and experienced initial improvement. Subsequently, she developed transient blurred vision in the

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right eye and right arm paresthesia that evolved with sudden brachiocrural paralysis and facial palsy. On physical examination, she was found to have central facial palsy, right brachial plegia and anesthesia, right crural severe paresis, and mild dysarthria with an National Institutes of Health Stroke Scale score of 19. Given that head computed tomography (CT) performed on admission revealed no bleeding, she received intravenous thrombolytic therapy and was transferred to a third level private hospital.

She was admitted to the intensive care unit on arrival. Head CT showed mild hypodensity in the region of the left middle cerebral artery, and neck vessel vessels US was performed using a linear and phased array probe. There was no alteration in the proximal portion of the left ICA in the two-dimentional mode. However, pulsed wave Doppler revealed a high-resistance flow pattern compatible with distal stenosis (Fig. 1). Because it is impossible to evaluate the distal portion of the ICA, we inserted an endocavitary probe covered with a sterile sheath through the mouth and placed it on the posterior wall of the pharynx. This approach allowed visualization of the distal portion of the ICA. In the color Doppler mode, we observed a homogeneous hypoechoic image with an absence of Doppler signal adjacent to a false lumen thrombosis associated with tuning of the true lumen with low-flow velocities (Figs. 2 and 3). Subsequently, transcranial US was performed. We identified the anterior (ACA), middle (MCA), and posterior cerebral artery in color Doppler mode, and less signal was present in the left MCA. Pulsed wave Doppler revealed increased velocities compatible with significant stenosis at the proximal portions of the MCA and ACA that were associated with a postobstructive flow pattern in the distal portion of the MCA (Figs. 4-7).

Transorbital US identified the intracranial portion of the ICA in

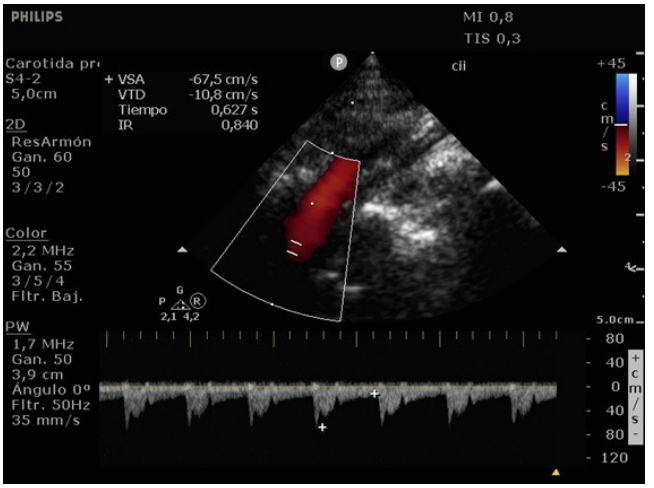


Fig. 1. Internal carotid artery conventional ultrasonography.

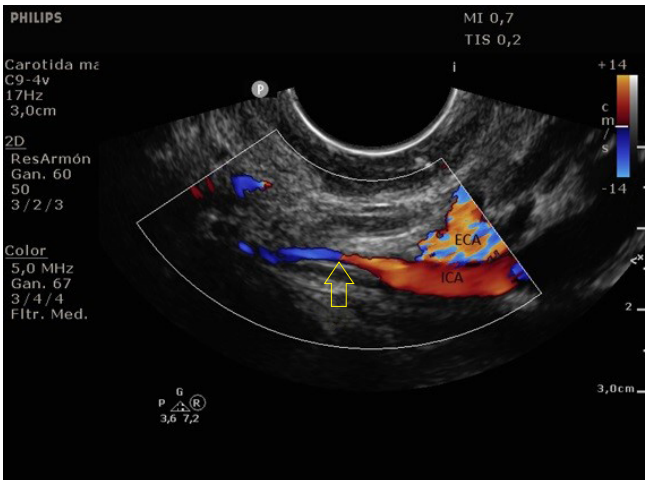


Fig. 2. Transoral ultrasonography. Yellow arrow, homogeneous hypoechoic image with absence of Doppler signal compatible with false lumen thrombosis and refinement of the true lumen.

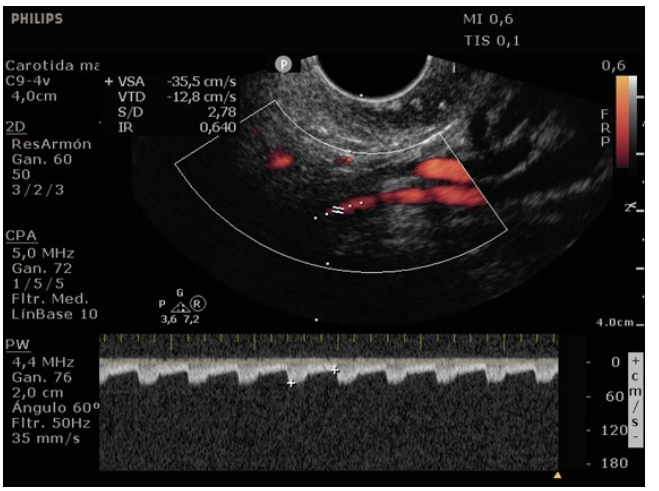


Fig. 3. Transoral ultrasonography using pulsed wave Doppler.

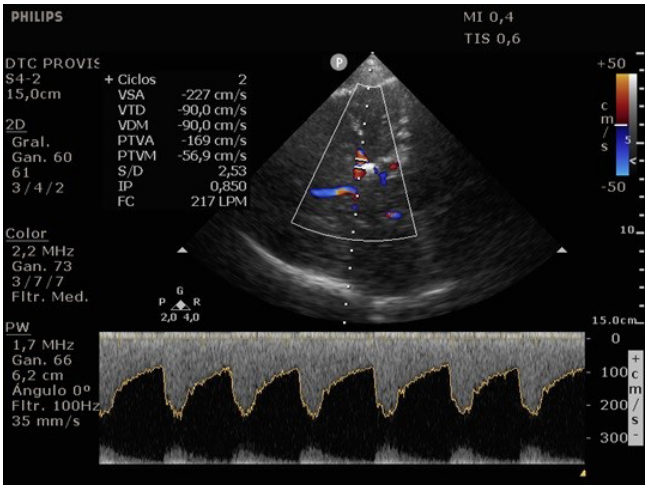


Fig. 4. Transcranial ultrasonography of the anterior cerebral artery.

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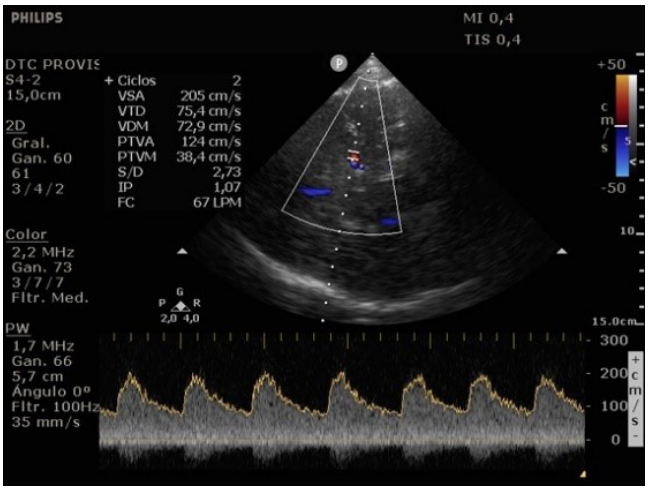


Fig. 5. Transcranial ultrasonography of the middle cerebral artery.

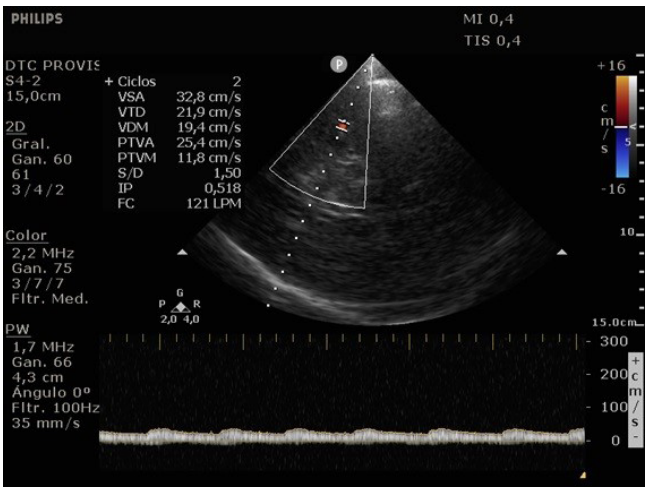


Fig. 6. Transcranial ultrasonography of the distal middle cerebral artery.

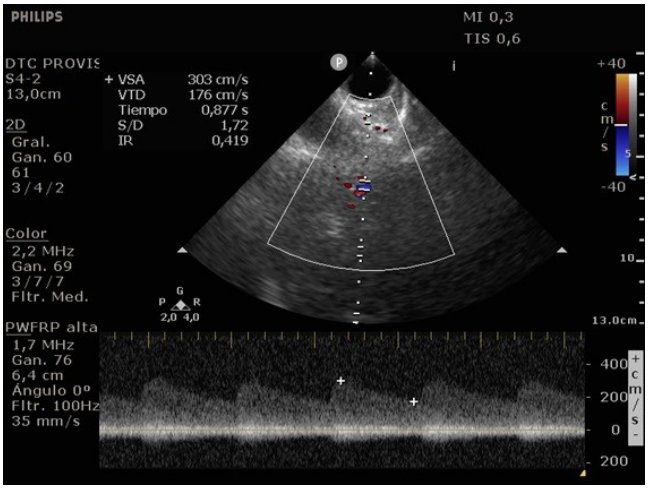


Fig. 7. Transorbital ultrasonography.

color Doppler mode, with the presence of very high velocities compatible with significant stenosis on the pulsed wave Doppler. Computed tomography angiography (CTA) of the neck and intracerebral vessels showed progressive lumen reduction at the distal portion of the left ICA until it became filiform and undetectable, along with the presence of flow in the intracranial arteries (Figs. 8 and 9). Brain magnetic resonance imaging detected positive diffusion-weighted imaging in the left MCA territory with hyperintensity at the left intracranial ICA in T2 compatible with the absence of emptied flow at the level of the petrous and cavernous portions of the artery (Fig. 10). The patient progressed with a

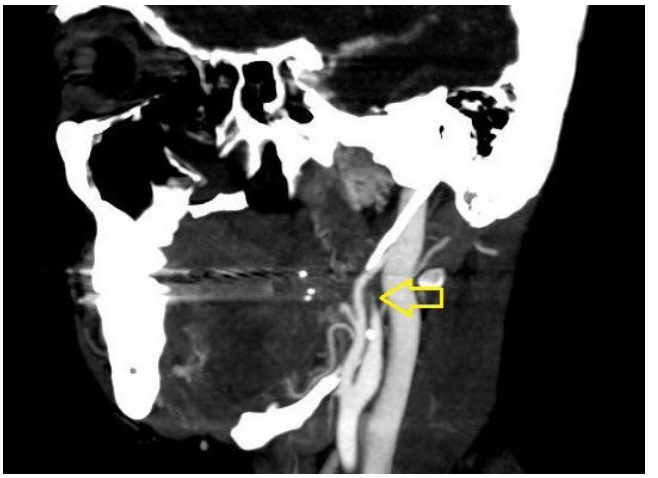


Fig. 8. Computed tomography angiography. Yellow arrow, postbulbar region of the left internal carotid artery (ICA) showing progressive reduction of the lumen until it becomes filiform and even undetectable compatible with ICA dissection.

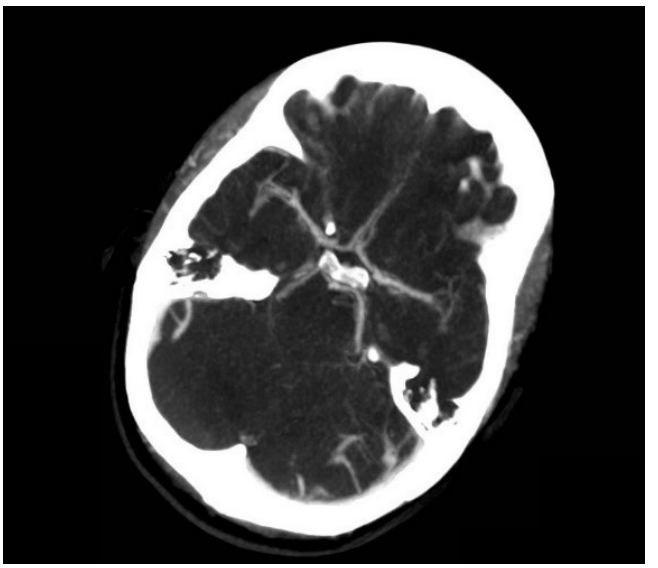


Fig. 9. Computed tomography angiography of the intracranial arteries.

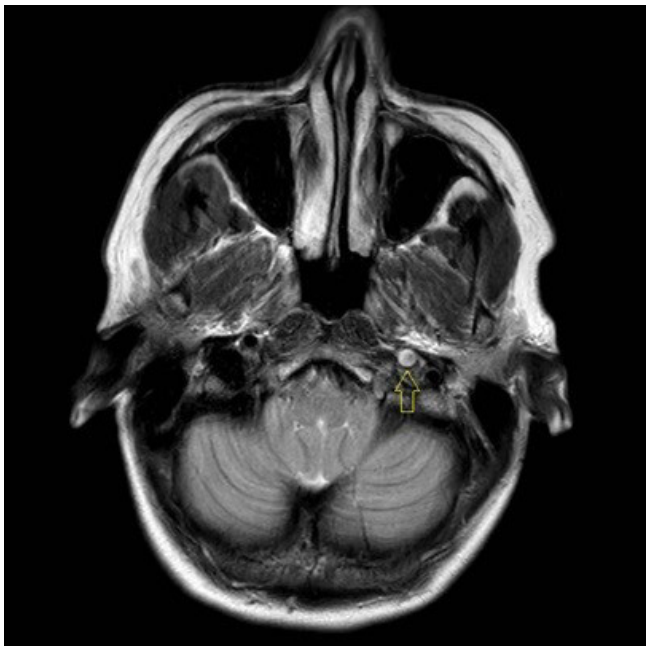


Fig. 10. Magnetic resonance imaging. Yellow arrow, absence of emptied flow at the level of the petrous and cavernous portion of the left internal carotid artery.

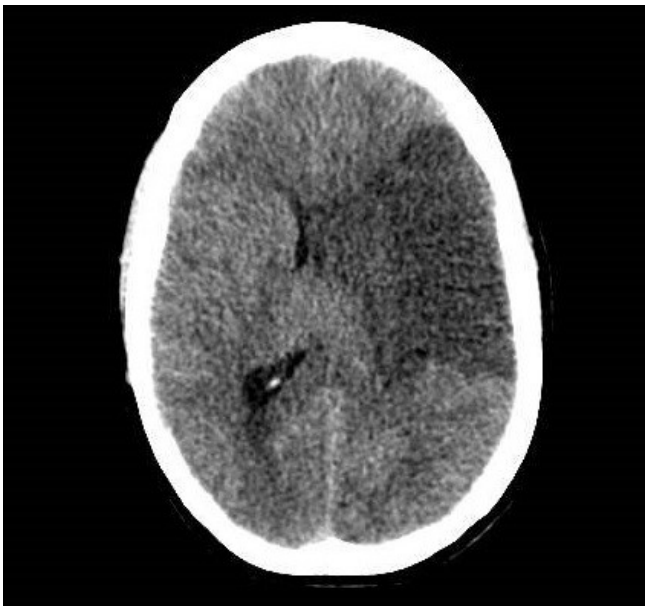


Fig. 11. Brain computed tomography of the patient in a coma.

Glasgow coma scale score of 8/15 and anisocoria with left mydriasis, requiring orotracheal intubation and mechanical ventilation support. We performed emergency head CT that showed extensive hypodensity in the left MCA territory with a 7-mm middle line shift (Fig. 11). We treated the patient with a hypertonic solution with partial reversal of mydriasis, and the neurosurgery team performed a left temporoparietal descompressive craniectomy. Despite surgical intervention, the patient developed mydriatic fixed pupils and lost the brain-steam reflexes. She was declared brain-dead in accordance with the criteria of the Argentinian death brain guidelines.

DISCUSSION

ICA dissection occurs when the intima is damaged and dissects the ICA walls with the formation of a hematoma. It causes stenosis, occlusion, and thrombus embolization. The etiology can be traumatic or spontaneous and is frequently related to structural abnormalities, such as collagen disease. There are four primary diagnostic modalities: digital subtraction angiography, which is the gold standard, magnetic resonance angiography (MRA), CTA, and US. The latter is frequently performed during the initial assessment and serial imaging. However, conventional US cannot assess the distal portion of the ICA because the maxillary bone interferes [3-5]. The convex and sectorial probes can be used to overcome this difficulty, but the image quality in B mode when these probes are used is suboptimal. As the ICA ascends vertically behind the posterolateral pharyngeal wall, it is possible to evaluate it using a transoral approach with an endocavitary probe [3]. With this method, ICA dissection is visualized as a refinement in color Doppler mode, with low velocities in pulsed wave Doppler, as evidenced in our case [6].

Suzuki et al. [7] published a series of cases demonstrating the usefulness of transoral US in the diagnosis of ICA dissection, which is comparable to that of digital subtraction angiography and MRA. Benninger et al. [8] reported the usefulness of the combination of the assessment or neck and intracerebral vessels using US in the diagnosis of this entity, based mainly on identifying altered arterial flow patterns. Similar results were reported by Wang et al. [9] in the ultrasonographic assessment of patients with stroke.

In our case report, the transoral ultrasound approach identified ICA dissection. Transcranial Doppler showed increased velocities in the proximal portions of the MCA and ACA and intracranial portion of the ICA that were compatible with significant stenosis. This finding was interpreted as dissection of the distal ICA that extended to the intracranial portion, as suspected due to the results of brain MRA until its bifurcation in the MCA and ACA.

In the present case, we report the usefulness of ICA US using both conventional and transoral approaches with transcranial evaluation as a noninvasive method for the diagnosis of dissection of the distal extra and intracranial portion of the ICA, as well as

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the involvement of the MCA in a patient with malignant cerebral infarction.

ARTICLE INFORMATION

Ethics statement

Approval for this study was waived in accordance with the local regulations because this study is a case report of a single patient and did not include protected health information, data analysis, or testing of a hypothesis, and was de-identified. Written consent was obtained from the patient before the publication of this case report.

Conflict of interest

No potential conflict of interest relevant to this article.

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Author contributions

Conceptualization: IC. Formal analysis: IC. Investigation: IC. Supervision: FMT. Writing–original draft: IC. Writing–review & editing: IC, VOC.

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